Exercises

Single Correct Answer Type

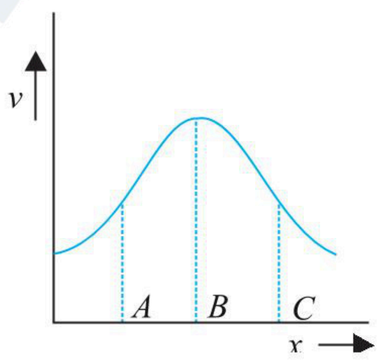
1. Mark the correct statement:
(1) If E is zero at a certain point, then V should be zero at that point.
(2) If E is not zero at a certain point, then V should not be zero at that point.
(3) If V is zero at a certain point, then E should be zero at that point.
(4) If V is zero at a certain point, then E may or may not be zero.

2. An uncharged conductor A is brought near a positively charged conductor B. Then
(1) the charge on B will increase, but the potential of B will not change
(2) the charge on B will not change, but the potential of B will decrease
(3) the charge on B will decrease, but the potential of B will not change
(4) the charge on B will not change, but the potential of B will increase

3. The electric field lines are closer together near object A than they are near object B. We can conclude that
(1) the potential near A is greater than the potential near B
(2) the potential near A is less than the potential near B
(3) the potential near A is equal to the potential near B
(4) nothing about the relative potentials near A and B

4. Mark the correct statement:
(1) An electron and a proton when released from rest in a uniform electric field experience the same force and the same acceleration.
(2) Two equipotential surfaces may intersect.
(3) A solid conducting sphere holds more charge than a hollow conducting sphere of the same radius.
(4) No work is done in taking a positive charge from one point to another inside a negatively charged metallic sphere.

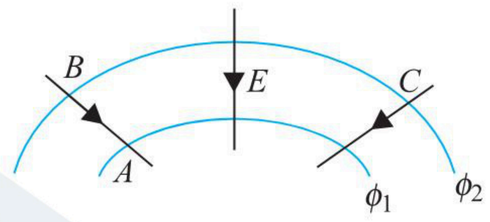
5. Variation of electrostatic potential along the x-direction is shown in figure. The correct statement about electric field is



(1) x-component at point B is maximum
(2) x-component at point A is toward positive x-axis
(3) x-component at point C is along negative x-axis
(4) x-component at point C is along positive x-axis

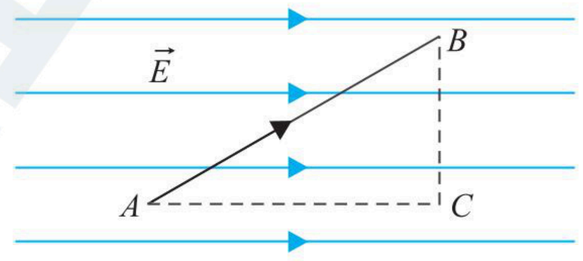
6. When the positively charged hanging pendulum bob is made fixed, the work done in slowly shifting a unit positive charge from infinity to P is V. If the pendulum is free to move, the corresponding work done is V'. Then
(1) $V = V'$ (2) $V > V'$
(3) $V < V'$ (4) $V \leq V'$

7. In moving from A to B along an electric field line, the work done by the electric field on an electron is 6.4×10^{-19} J. If ϕ_1 and ϕ_2 are equipotential surfaces, then the potential difference $V_C - V_A$ is



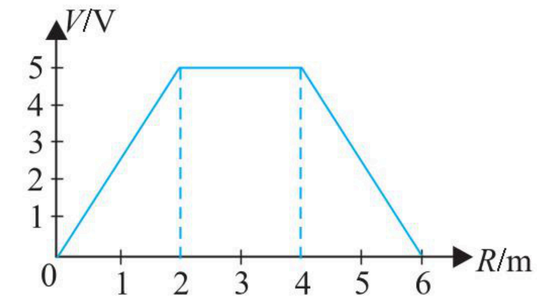
(1) –4 V (2) 4 V
(3) zero (4) 6.4 V

8. An electron is taken from point A to point B along the path AB in a uniform electric field of intensity E = 10 Vm^{-1}. Side AB = 5 m, and side BC = 3 m.



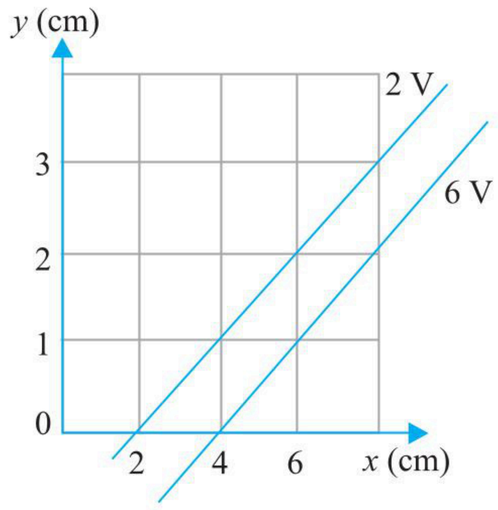
Then, the amount of work done on the electron by us is
(1) 50 eV (2) 40 eV
(3) –50 eV (4) –40 eV

9. The variation of potential with distance R from the fixed point is shown in figure.



The electric field at R = 5 m is
(1) 2.5 Vm^{-1} (2) –2.5 Vm^{-1}
(3) 0.4 Vm^{-1} (4) –0.4 Vm^{-1}

10. Figure shows two equipotential lines in the xy plane for an electric field. The scales are marked. The x-component and y-component of the field in the space between these equipotential lines are, respectively



(1) +100 Vm^{-1}, –200 Vm^{-1}
(2) –100 Vm^{-1}, +200 Vm^{-1}
(3) +200 Vm^{-1}, +100 Vm^{-1}
(4) –200 Vm^{-1}, –400 Vm^{-1}

11. A large insulated sphere of radius r charged with Q units of electricity is placed in contact with a small insulated uncharged sphere of radius r' and is then separated. The charge on the smaller sphere will now be
(1) $\frac{Q(r'+r)}{r'}$ (2) $\frac{Q(r'+r)}{r}$
(3) $\frac{Qr}{r'+r}$ (4) $\frac{Qr'}{r'+r}$

12. When a 2 μC charge is carried from point A to point B, the amount of work done by the electric field is 50 μJ. What is the potential difference and which point is at a higher potential?

(1) 25 V, B (2) 25 V, A

(3) 20 V, B (4) both are at same potential

13. An electric field is expressed as $\vec{E} = 2\hat{i} + 3\hat{j}$. Find the potential difference $(V_A - V_B)$ between two points A and B whose position vectors are given by $r_A = \hat{i} + 2\hat{j}$ and $r_B = 2\hat{i} + \hat{j} + 3\hat{k}$.

(1) −1 V (2) 1 V

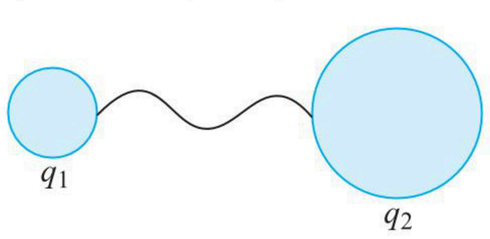
(3) 2 V (4) 3 V

14. The potential function of an electrostatic field is given by $V = 2x^2$. Determine the electric field strength at the point (2 m, 0, 3 m).

(1) $\vec{E} = 4\hat{i}\ (\text{NC}^{-1})$ (2) $\vec{E} = -4\hat{i}\ (\text{NC}^{-1})$

(3) $\vec{E} = 8\hat{i}\ (\text{NC}^{-1})$ (4) $\vec{E} = -8\hat{i}\ (\text{NC}^{-1})$

15. Two spherical conductors of radii R_1 and R_2 are separated by a distance much larger than the radius of either sphere. The spheres are connected by a conducting wire as shown in figure. If the charges on the spheres in equilibrium are q_1 and q_2, respectively, what is the ratio of the field strength at the surfaces of the spheres?



(1) R_2/R_1 (2) R_2^2 / R_1^2

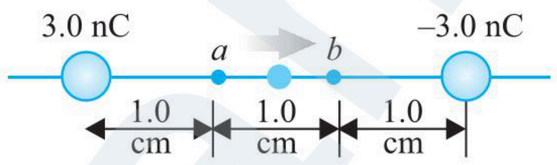
(3) R_1 / R_2 (4) R_1^2 / R_2^2

16. There is an electric field E in the x-direction. If the work done by the electric field in moving a charge of 0.2 C through a distance of 2 m along a line making an angle 60° with the x-axis is 4 J, then what is the value of E?

(1) $\sqrt{3}\ \text{NC}^{-1}$ (2) $4\ \text{NC}^{-1}$

(3) $5\ \text{NC}^{-1}$ (4) $20\ \text{NC}^{-1}$

17. As shown in figure, a dust particle with mass $m = 5.0 \times 10^{-9}$ kg and charge $q_0 = 2.0$ nC starts from rest at point a and moves in a straight line to point b. What is its speed v at point b?



(1) 26 ms^{-1} (2) 34 ms^{-1}

(3) 46 ms^{-1} (4) 14 ms^{-1}

18. Two charged particles having charges 1 and −1 μC and of mass 50 g each are held at rest while their separation is 2 m. Now the charges are released. Find the speed of the particles when their separation is 1 m.

(1) $\frac{1}{5}\text{ms}^{-1}$ (2) $\frac{3}{5}\text{ms}^{-1}$

(3) $\frac{3}{10}\text{ms}^{-1}$ (4) $\frac{2}{7}\text{ms}^{-1}$

19. A 100 eV electron is projected directly toward a large metal plate that has surface charge density of $-2.0 \times 10^{-6}\ \text{Cm}^{-2}$. From what distance must the electron be projected, if it is to just fail to strike that plate?

(1) 0.44 mm (2) 0.20 mm

(3) 1 mm (4) 0.30 mm

20. A point charge q is placed inside a conducting spherical shell of inner radius $2R$ and outer radius $3R$ at a distance of R from the center of the shell. Find the electric potential at the center of the shell.

(1) $\frac{1}{4\pi\varepsilon_0}\frac{q}{2R}$ (2) $\frac{1}{4\pi\varepsilon_0}\frac{4q}{3R}$

(3) $\frac{1}{4\pi\varepsilon_0}\frac{5q}{6R}$ (4) $\frac{1}{4\pi\varepsilon_0}\frac{2q}{3R}$

21. n charged drops, each of radius r and charge q, coalesce to form a big drop of radius R and charge Q. If V is the electric potential and E is the electric field at the surface of a drop, then

(1) $E_{\text{big}} = n^{2/3}E_{\text{small}}$ (2) $V_{\text{big}} = n^{1/3}V_{\text{small}}$

(3) $E_{\text{small}} = n^{2/3}E_{\text{big}}$ (4) $V_{\text{big}} = n^{2/3}V_{\text{small}}$

22. Four identical charges are placed at the points (1, 0, 0), (0, 1, 0), (−1, 0, 0), and (0, −1, 0). Then,

(1) the potential at the origin is zero

(2) the electric field at the origin is not zero

(3) the potential at all points on the z-axis, other than the origin, is zero

(4) the field at all points on the z-axis, other than the origin, acts along the z-axis

23. Two point charges Q and $-Q/4$ placed along the x-axis are separated by a distance r. Take $-Q/4$ as origin and it is placed at the right of Q. Then the potential is zero

(1) at $x = r/3$ only

(2) at $x = -r/5$ only

(3) both at $x = r/3$ and at $x = -r/5$

(4) there exist two points on the axis where the electric field is zero

24. The electric potential decreases uniformly from 120 V to 80 V as one moves on the x-axis from $x = -1$ cm to $x = +1$ cm. The electric field at the origin

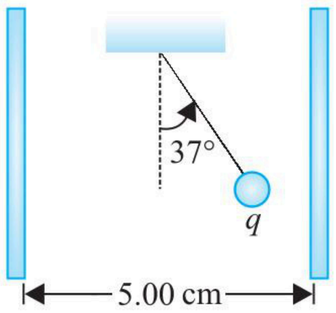
(1) must be equal to 20 Vcm^{-1}

(2) must be equal to 20 Vm^{-1}

(3) may be greater than 20 Vcm^{-1}

(4) may be less than 20 Vcm^{-1}

25. A small sphere with mass 1.2 g hangs by a thread between two parallel vertical plates 5.00 cm apart. The plates are insulating and have uniform surface charge densities $+\sigma$ and $-\sigma$. The charge on the sphere is $q = 9 \times 10^{-6}$ C. What potential difference between the plates will cause the thread to assume an angle of 37° with the vertical as shown in figure?



(1) 30 V (2) 12 V

(3) 50 V (4) 25 V

26. A particle of mass m carrying charge q is projected with velocity v from point P toward an infinite line of charge from a distance a. Its speed reduces to zero momentarily at point Q, which is at a distance $a/2$ from the line of charge. If another particle with mass m and charge $-q$ is projected with the same velocity v from P toward the line of charge, what will be its speed at Q?

(1) $\sqrt{2}v$ (2) $2v$

(3) $\sqrt{v}$ (4) $\sqrt{2}$

27. A solid conducting sphere of radius 10 cm is enclosed by a thin metallic shell of radius 20 cm. A charge $q = 20\ \mu C$ is given to the inner sphere. Find the heat generated in the process when the inner sphere is connected to the shell by a conducting wire.

(1) 12 J (2) 9 J

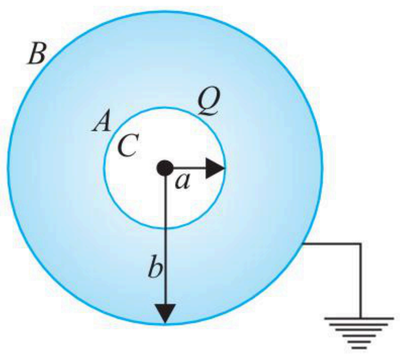
(3) 24 J (4) zero

28. Find the potential V of an electrostatic field $\vec{E} = a(y\hat{i} + x\hat{j})$, where a is a constant.

(1) $axy + C$ (2) $-axy + C$

(3) axy (4) $-axy$

29. A conducting sphere A of radius a, with charge Q, is placed concentrically inside a conducting shell B of radius b. B is earthed. C is the common center of A and B. Study the following statements.



(i) The potential at a distance r from C, where $a \le r \le b$, is $\frac{1}{4\pi\varepsilon_0}\left(\frac{Q}{r}\right)$

(ii) The potential difference between A and B is $\frac{1}{4\pi\varepsilon_0}Q\left(\frac{1}{a} - \frac{1}{b}\right)$

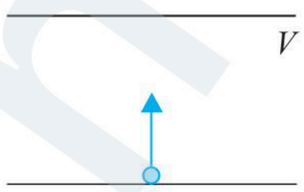
(iii) The potential at a distance r from C, where $a \le r \le b$, is $\frac{1}{4\pi\varepsilon_0}Q\left(\frac{1}{r} - \frac{1}{b}\right)$

Which of the following statements are correct?

(1) Only (i) and (ii) (2) Only (ii) and (iii)

(3) Only (i) and (iii) (4) All

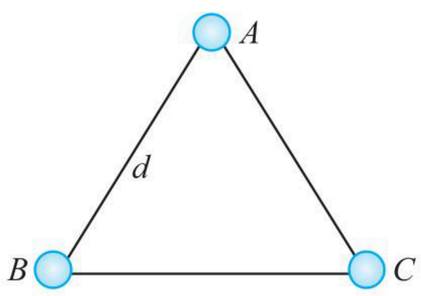
30. An electron having charge e and mass m starts from the lower plate of two metallic plates separated by a distance d. If the potential difference between the plates is V, the time taken by the electron to reach the upper plate is given by



(1) $\sqrt{\frac{2md^2}{eV}}$ (2) $\sqrt{\frac{md^2}{eV}}$

(3) $\sqrt{\frac{md^2}{2eV}}$ (4) $\frac{2md^2}{eV}$

31. Three identical metallic uncharged spheres A, B, and C, each of radius a, are kept at the corners of an equilateral triangle of side d ($d >> a$) as shown in figure. The fourth sphere (of radius a), which has a charge q, touches A and is then removed to a position far away. B is earthed and then the earth connection is removed. C is then earthed. The charge on C is



(1) $\frac{qa}{2d}\left(\frac{2d-a}{2d}\right)$ (2) $\frac{qa}{2d}\left(\frac{2d-a}{d}\right)$

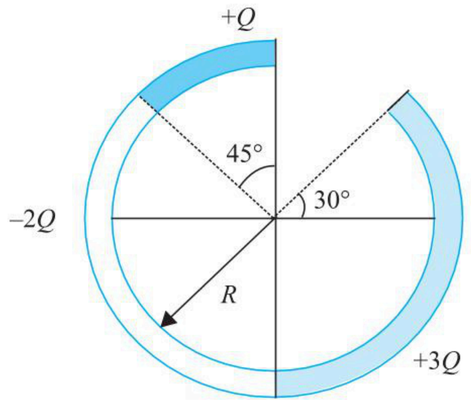
(3) $-\frac{qa}{2d}\left(\frac{d-a}{d}\right)$ (4) $\frac{2qa}{d}\left(\frac{d-a}{2d}\right)$

32. Two concentric conducting spherical shells of radii a_1 and a_2 ($a_2 > a_1$) are charged to potentials V_1 and V_2, respectively. Find the charge on the inner shell.

(1) $q_1 = 4\pi\varepsilon_0\left(\frac{V_1 - V_2}{a_2 - a_1}\right)a_1a_2$ (2) $q_1 = 4\pi\varepsilon_0\left(\frac{V_1 + V_2}{a_2 + a_1}\right)a_1a_2$

(3) $q_1 = \pi\varepsilon_0\left(\frac{V_1 - V_2}{a_2 + a_1}\right)a_1a_2$ (4) $q_1 = 2\pi\varepsilon_0\left(\frac{V_1 + V_2}{a_2 - a_1}\right)a_1a_2$

33. Figure shows three circular arcs, each of radius R and total charge as indicated. The net electric potential at the center of curvature is



(1) $\frac{Q}{2\pi\varepsilon_0 R}$

(2) $\frac{5Q}{12\pi\varepsilon_0 R}$

(3) $\frac{3Q}{32\pi\varepsilon_0 R}$

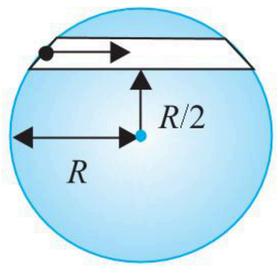
(4) none of these

34. The point charge q is within the cavity of an electrically neutral conducting shell whose outer surface has spherical shape. Find the potential V at a point P lying outside the shell at a distance r from the center O of the outer sphere.

(1) $V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$ (2) $V = \frac{1}{2\pi\varepsilon_0}\frac{q}{r}$

(3) $V = \frac{1}{\pi\varepsilon_0}\frac{q}{r}$ (4) $V = \frac{3}{4\pi\varepsilon_0}\frac{q}{r}$

35. A unit positive point charge of mass m is projected with a velocity V inside the tunnel as shown. The tunnel has been made inside a uniformly charged nonconducting sphere. The minimum velocity with which the point charge should be projected such that it can reach the opposite end of the tunnel is equal to



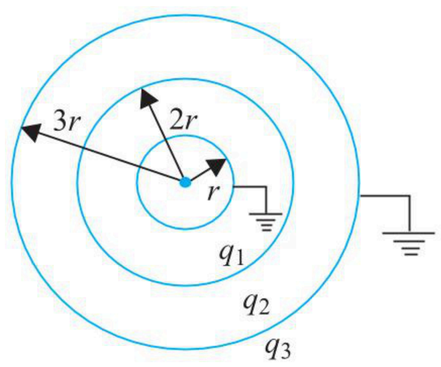
(1) $[\rho R^2/4m\varepsilon_0]^{1/2}$

(2) $[\rho R^2/24m\varepsilon_0]^{1/2}$

(3) $[\rho R^2/6m\varepsilon_0]^{1/2}$

(4) zero because the initial and the final points are at same potential

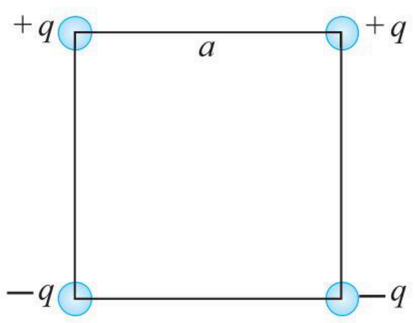
36. Three concentric conducting spherical shells have radii r, $2r$, and $3r$ and charges q_1, q_2, and q_3, respectively. Innermost and outermost shells are earthed as shown in figure. The charges shown are after earthing. Select the correct alternative.



(1) $q_1 + q_3 = -q_2$ (2) $q_1 = -q_2$

(3) $\frac{q_3}{q_2} = -\frac{1}{3}$ (4) None of these

37. Four charges are placed at four corners of a square as shown figure. The side of the square is a. Two charges are positive and two are negative, but their magnitudes are the same. Now, an external agent starts decreasing all the sides of the square slowly and at the same rate. What happens to the electrical potential energy of the system and what will be the nature of work done by the agent?

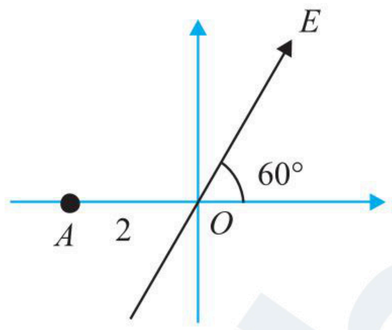


(1) increases, positive (2) increases, negative
(3) decreases, negative (4) decreases, positive

38. At a distance r from a point located at origin in space, the electric potential varies as $V = 10r$. Find the electric field at $\vec{r} = 3\hat{i} + 4\hat{j} - 5\hat{k}$.

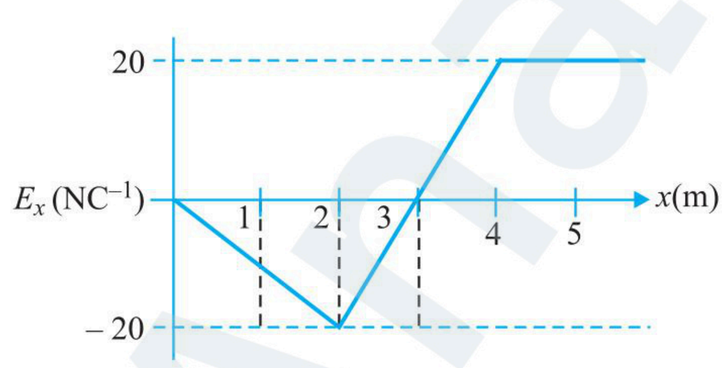
(1) $\sqrt{2}(3\hat{i} + 4\hat{j} - 5\hat{k})$ (2) $-\sqrt{2}(3\hat{i} + 4\hat{j} - 5\hat{k})$
(3) $-\sqrt{3}(3\hat{i} + 4\hat{j} - 5\hat{k})$ (4) None of these

39. A uniform electric field of 100 Vm^{-1} is directed at 60° with the positive x-axis as shown in figure. If $OA = 2$ m, the potential difference $V_O - V_A$ is



(1) –50 V
(2) 50 V
(3) 100 V
(4) –100 V

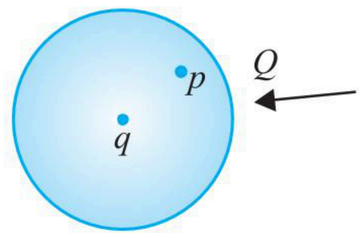
40. A graph of the x-component of the electric field as a function of x in a region of space is shown in figure. The y- and z-components of the electric field are zero in this region. If the electric potential is 10 V at the origin, then the potential at $x = 2.0$ m is



(1) 10 V (2) 40 V
(3) –10 V (4) 30 V

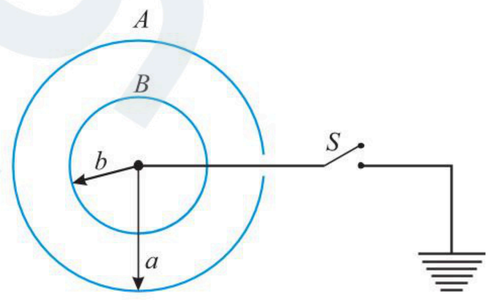
41. An uncharged conductor A is brought near a positively charged conductor B. The size of the conductor A is much greater than the size of conductor B. Then,
(1) the charge on B will increase, but the potential of B will not change
(2) the charge on B will not change, but the potential of B will decrease
(3) the charge on B will decrease, but the potential of B will not change
(4) the charge on B will not change, but the potential of B will increase

42. Inside a hollow conducting sphere, which is uncharged, a charge q is placed at its center. Let electric field at a distance x from center at point p be E and potential at this point be V. Now, some positive charge Q is given to this sphere, then



(1) E will remain the same (2) E will increase
(3) V will decrease (4) V will remain the same

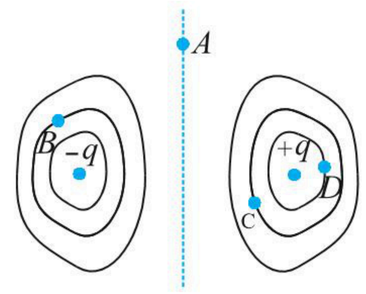
43. Consider two concentric metal spheres. The outer sphere is given a charge $Q > 0$, then



(1) the electrons will flow from earth to inner sphere if S is shorted
(2) the electrons will flow from inner sphere to the earth if S is shorted
(3) the shorting of S will produce a charge of $-Q$ on the inner sphere
(4) none of these

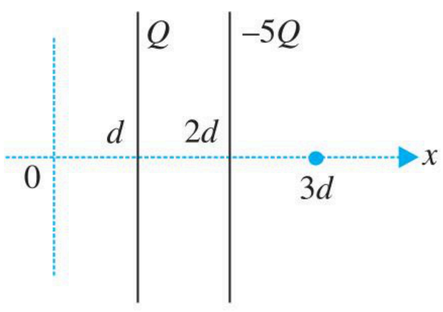
44. Two point charges $+Q$ each have been placed at the positions $(-a/2, 0, 0)$ and $(a/2, 0, 0)$. The locus of the points where $-Q$ charge can be placed such that total electrostatic potential energy of the system can become equal to zero, can be represented by which of the following equations?
(1) $z^2 + (y - a)^2 = 2a$ (2) $z^2 + (y - a)^2 = 27a^2/4$
(3) $z^2 + y^2 = 15a^2/4$ (4) None of these

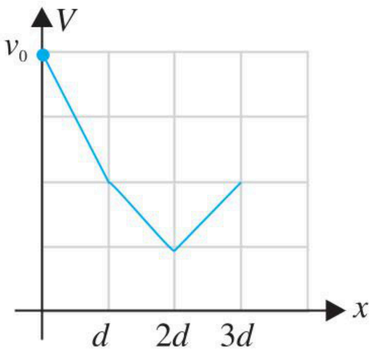
45. Figure shows equipotential surfaces for a two charges system. At which of the labeled points will an electron have the highest potential energy?



(1) point A
(2) point B
(3) point C
(4) point D

46. Two large identical plates are placed in front of each other at $x = d$ and $x = 2d$ as shown in the figure. If charges on plates are Q and $-5Q$, the potential versus distance graph for region $x = 0$ to $x = 3d$ is (d is very small and potential at $x = 0$ is v_0)



(1)


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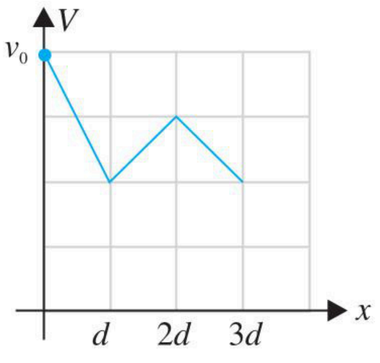

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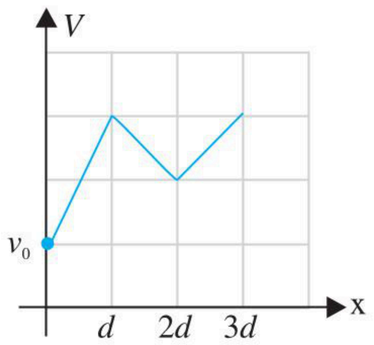

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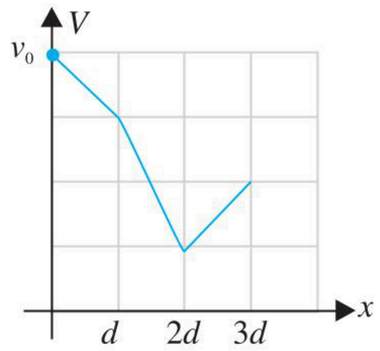
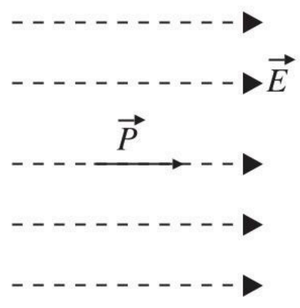
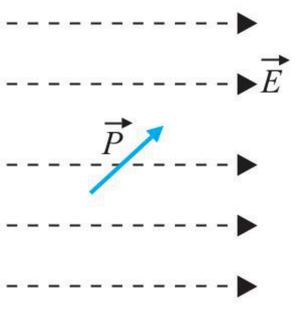


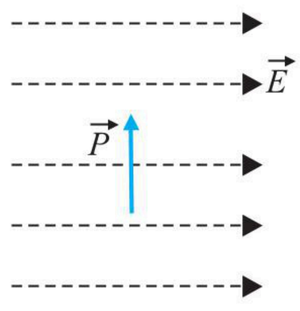
47. An electric dipole of dipole moment $\vec{p}$ is oriented parallel to a uniform electric field $\vec{E}$, as shown. It is rotated to one of the four orientations shown below. Rank the final orientations according to the change in the potential energy of the dipole–field system, most negative to most positive.



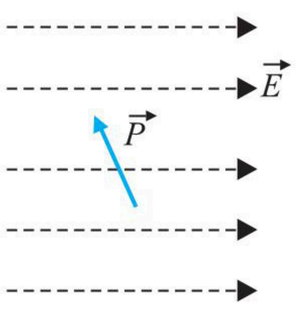
(i)



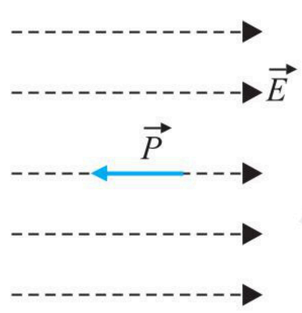
(ii)



(iii)

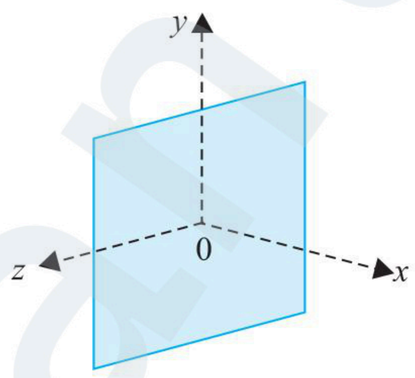


(iv)



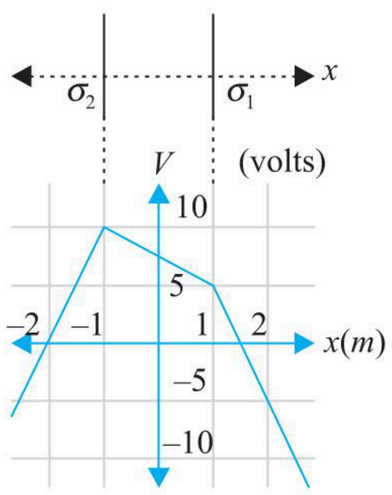
(1) i., ii., iv., iii.
(2) iv., iii., ii., i.
(3) i., ii., iii., iv.
(4) iii., ii., and iv. tie, then i.

48. A plane sheet has a uniformly distributed charge. Origin is selected at the middle of the sheet (in yz plane). A charge $+Q$ is moved from point A $(10a, 2a, 0)$ to point B $(10a, 0, 2a)$ along a circle and then to point C $(10a, 0, 0)$. If the initial electrostatic potential energy is U_0, what will be final electrostatic potential energy?



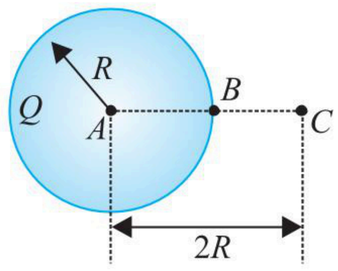
(1) zero
(2) U_0
(3) $U_0\sqrt{2}$
(4) $\frac{U_0}{2}$

49. Figure shows the potential due to two similarly charged infinite sheets with charge per unit area σ_2 and σ_1. From examining this plot we can deduce that



(1) $\sigma_2 > \sigma_1$
(2) $\sigma_2 < \sigma_1$
(3) $\sigma_2 = \sigma_1$
(4) None of these

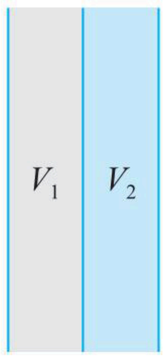
50. Find the ratio of electric work done in bringing a charge q from A to B (W_{AB}) and then from B to C (W_{BC}) in a sphere of charge Q distributed uniformly throughout its volume. $(W_{AB}/W_{BC} = ?)$



(1) 1
(2) 1.5
(3) 0.75
(4) None of these

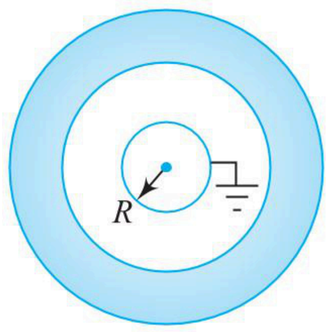
51. A nonuniformly charged ring is kept near an uncharged conducting solid sphere. The distance between their centres (which are on the same line normal to the plane of the ring) is 3 m and their radius is 4 m. If total charge on the ring is 1 μC, then the potential of the sphere will be
(1) 2.25 kV
(2) 3 kV
(3) 1.8 kV
(4) cannot be determined

52. An electron traveling in a uniform electric field passes from a region of potential V_1 to a region of higher potential V_2. Then



(1) no change takes place in velocity component parallel to interface of two regions
(2) direction of its motion remains unchanged but speed increases
(3) direction of its motion may change but speed must be decreased
(4) decrease in kinetic energy is proportional to $\sqrt{V_2 - V_1}$

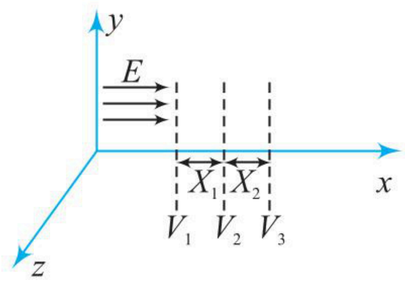
53. A conducting sphere of radius R and a concentric thick spherical shell of inner radius $2R$ and outer radius $3R$ is shown in figure. A charge $+10Q$ is given to the shell and inner sphere is earthed. Then charge on inner sphere is



(1) $-4Q$
(2) $-10Q$
(3) zero
(4) none of these

54. Let V_0 be the potential at the origin in an electric field $\vec{E} = 4\hat{i} + 5\hat{j}$. The potential at the point (x, y) is
(1) $V_0 - 4x - 5y$
(2) $V_0 + 4x + 5y$
(3) $4x + 5y - V_0$
(4) $-4x - 5y$

55. In an electric field shown in figure, three equipotential surfaces are shown. If function of electric field is $E = 2x^2$ Vm^{-1}, and given that $V_1 - V_2 = V_2 - V_3$, then we have



(1) $x_1 = x_2$
(2) $x_1 > x_2$
(3) $x_2 > x_1$
(4) data insufficient

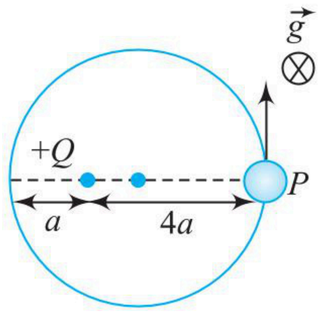
56. The potential of a point B (–20 m, 30 m) taking the potential of a point A (30 m, –20 m) to be zero in an electric field $\vec{E} = 10x\hat{i} - 20\hat{j}$ NC^{-1} is
(1) 350 V
(2) –100 V
(3) 300 V
(4) 3500 V

57. A positive charge $+q_1$ is located to the left of a negative charge $-q_2$. On a line passing through the two charges, there are two places where the total potential is zero. The reference is assumed to be at infinity. The first place is between the charges and is 4.00 cm to the left of the negative charge. The second place is 7.00 cm to the right of the negative charge. If $q_2 = -12$ μC, what is the value of charge q_1.

(1) 44 μC (2) 12 μC

(3) 32 μC (4) 64 μC

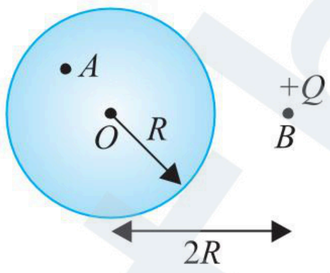
58. The diagram shows a small bead of mass m carrying charge q. The bead can freely move on the smooth fixed ring placed on a smooth horizontal plane. In the same plane a charge $+Q$ has also been fixed as shown. The potential at the point P due to $+Q$ is V. The velocity with which the bead should projected from the point P so that is can complete a circle should be greater than



(1) $\sqrt{\frac{6qV}{m}}$ (2) $4\sqrt{\frac{qV}{m}}$

(3) $\sqrt{\frac{3qV}{m}}$ (4) $\sqrt{\frac{7qV}{2m}}$

59. A point charge $+Q$ is placed at point B at a distance $2R$ from the center O of an uncharged thin conducting shell of radius R as shown in the figure. If V_A be the potential at point A, which is at a radial distance of $R/2$ from the center of the shell, then



(1) $V_A > \frac{Q}{8\pi\varepsilon_0 R}$ (2) $V_A < \frac{Q}{8\pi\varepsilon_0 R}$

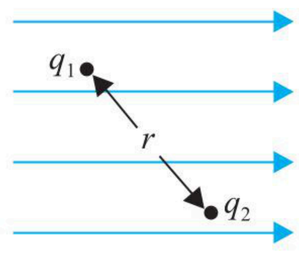
(3) $V_A = \frac{Q}{8\pi\varepsilon_0 R}$ (4) $V_A = 0$

60. Uniform electric field exists in a region and is given by $\vec{E} = E_0\hat{i} + E_0\hat{j}$. There are four points $A(-a, 0)$, $B(0, -a)$, $C(a, 0)$, and $D(0, a)$ in the xy plane. Which of the following is the correct relation for the electric potential?

(1) $V_A = V_C > V_B = V_D$ (2) $V_A = V_B > V_C = V_D$

(3) $V_A > V_C > V_B = V_D$ (4) $V_A < V_C < V_B < V_D$

61. Two point charges q_1 and q_2 are placed in an external uniform electric field as shown in figure. The potential at the location of q_1 and q_2 are V_1 and V_2, respectively. The effect of q_1 and q_2 is not included in V_1 and V_2, i.e., V_1 and V_2 are potentials at location of q_1 and q_2 due to external unspecified charges only. Then electric potential energy for this configuration of two charged particle is



(1) $\frac{q_1V_1 + q_2V_2}{2}$ (2) $q_1V_1 + q_2V_2$

(3) $q_1V_1 + q_2V_2 + \frac{q_1q_2}{4\pi\varepsilon_0 r}$ (4) $\frac{q_1q_2}{4\pi\varepsilon_0 r}$

62. There are two uncharged identical metallic spheres 1 and 2 of radius r separated by a distance d ($d >> r$). A charged metallic sphere of same radius having charge q is touched with one of the sphere. After some time it is moved away from the system. Now the uncharged sphere is earthed. Charge on earthed sphere is

(1) $q/2$ (2) $-qr/d$

(3) $-qr/2d$ (4) $-q/2$

63. Eight point charges of charge q each are placed on the eight corners of a cube of side a. A solid neutral metallic sphere of radius $a/3$ is placed with its center at the center of the cube. As a result, charges are induced on the sphere, which form certain patterns on its surface. What is the potential of the sphere?

(1) $\frac{8}{\sqrt{3}}\frac{kq}{a}$ (2) $\frac{16}{\sqrt{3}}\frac{kq}{a}$

(3) $\frac{-8}{\sqrt{3}}\frac{kq}{a}$ (4) zero

Multiple Correct Answers Type

1. Two point charges of different magnitudes and of opposite signs are separated by some distance. There can be

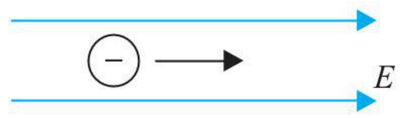
(1) only one point in space where net electric field intensity is zero

(2) only two points in space where net electric potential is zero

(3) infinite number of points in space where net electric field intensity is 0

(4) infinite number of points in space where net electric potential is zero

2. A negative charge is moved by an external agent in the direction of electric field. Then



(1) potential energy of the charge increases

(2) potential energy of the charge decreases

(3) positive work is done by the electric field

(4) negative work is done by the electric field

3. If a charged conductor is enclosed by a hollow charged conducting shell (assumed concentric and spherical in shape), and they are connected by a conducting wire, then which of the following statement(s) would be correct?

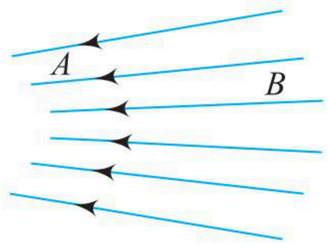
(1) Potential difference between the two conductors becomes zero.

(2) If charge on the inner conductor is q and on the outer conductor is $2q$, then finally charge on the outer conductor will be $3q$.

(3) The charge on the inner conductor is totally transferred to the outer conductor.

(4) If charge on the inner conductor is q and charge on the outer conductor is zero, then finally the charge on each conductor will be $q/2$.

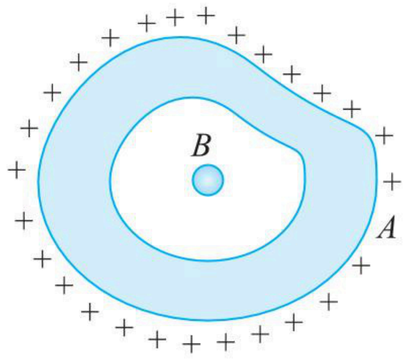
4. Which of the following is true for the figure showing electric lines of force? (E is electrical field, and V is potential.)



(1) $E_A > E_B$ (2) $E_B > E_A$

(3) $V_A > V_B$ (4) $V_B > V_A$

5. An uncharged conducting ball B lies inside a charged conductor A as shown in figure. B is isolated from A.



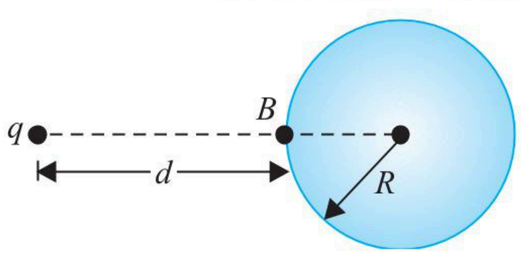
(1) When B touches the inner surface of A, then potential of B will change.

(2) No net charge is inside the cavity.

(3) There is an induced charge on B.

(4) Potentials of A and B are same.

6. For the situation shown in figure, mark the correct statement(s).



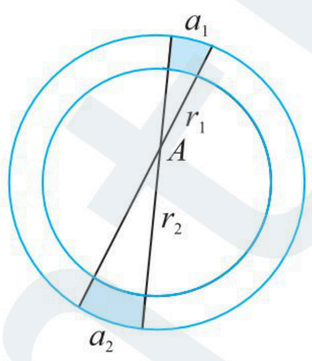
(1) Potential of the conductor is $q/[4\pi\varepsilon_0(d+R)]$.

(2) Potential of the conductor is $q/4\pi\varepsilon_0 d$.

(3) Potential of the conductor cannot be determined as the nature of distribution of induced charges is not known.

(4) Potential at point B due to the induced charges is $-qR/[4\pi\varepsilon_0(d+R)d]$.

7. A wire having a uniform linear charge density λ is bent in the form of a ring of radius R. Point A as shown in figure is in the plane of the ring but not at the center. Two elements of the ring of lengths a_1 and a_2 subtend very small same angle at point A. They are at distances r_1 and r_2 from point A, respectively ($r_2 > r_1$).



(1) The ratio of charges of element a_1 to that of element a_2 is r_1/r_2.

(2) The element a_1 produced greater magnitude of electric field at A than element a_2.

(3) The elements a_1 and a_2 produce same potential at A.

(4) The direction of the net electric field produced by the elements only at A is toward element a_2.

8. At a distance of 5 cm and 10 cm outward from the surface of a uniformly charged solid sphere, the potentials are 100 V and 75 V, respectively. Then

(1) potential at its surface is 150 V

(2) the charge on the sphere is $(5/3) \times 10^{-10}$ C

(3) the electric field on the surface is 1500 Vm^{-1}

(4) the electric potential at its center is 0 V

9. Two infinite, parallel, nonconducting sheets carry equal positive charge density σ. One is placed in the yz plane and the other at distance $x = a$. Take potential $V = 0$ at $x = 0$. Then

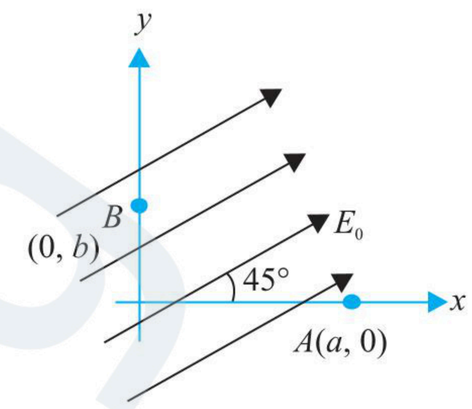
(1) for $0 \le x \le a$, potential $V = 0$

(2) for $x \ge a$, potential $V = \dfrac{\sigma}{\varepsilon_0}(x-a)$

(3) for $x \ge a$, potential $V = -\dfrac{\sigma}{\varepsilon_0}(x-a)$

(4) for $x \le 0$, potential $V = \dfrac{\sigma}{\varepsilon_0}x$

10. A uniform electric field E_0 exists in a region at angle 45° with the x-axis. There are two point A (a, 0) and B (0, b) having potential V_A and V_B, respectively, then



(1) $a < b$

(2) $a > b$

(3) $a = b$

(4) None of these

11. A small sphere is charged uniformly and placed at some point $A(x_0, y_0)$ so that at point B(9 m, 4 m) electric field strength is $\vec{E} = (54\hat{i} + 72\hat{j})$ NC^{-1} and potential is 1800 V. Then

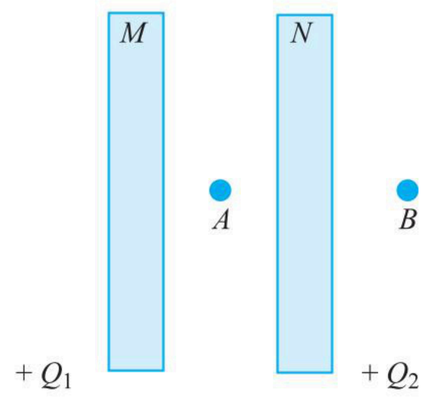
(1) the magnitude of charge on the sphere is 4 μC

(2) the magnitude of charge on the sphere is 2 μC

(3) coordinates of A are $x_0 = -3$ m, $y_0 = -12$ m

(4) coordinates of A are $x_0 = 4$ m, $y_0 = -1$ m

12. Two conducting plates M and N, each having large surface area A (on one side), are placed parallel to each other figure. The plate M is given charge Q_1 and N, charge Q_2 ($<Q_1$). Then



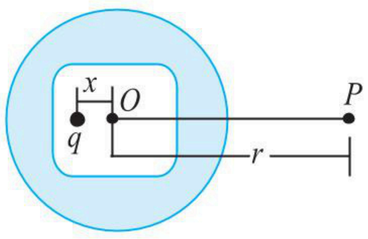
(1) electric field at point A is $(Q_1 - Q_2)/2A\varepsilon_0$ toward right

(2) electric field at point B is $(Q_1 + Q_2)/2A\varepsilon_0$ toward right

(3) electric potential of N is greater than M

(4) all of the above

13. A point charge q is placed within the cavity of an electrically neutral conducting shell whose outer surface has spherical shape figure. Then,



(1) the potential V at point P lying outside the shell at a distance r from the centre O of the outer surface depends upon the value of x

(2) potential at P does not depend upon the value of x

(3) a total charge q will be induced on the outer surface of the shell which will be distributed uniformly on the outer surface

(4) a total charge $-q$ will be induced on the inner surface of the shell, which will be distributed nonuniformly on the inner surface

14. The electric potential at a certain distance from a point charge is 600 V and the electric field is 200 NC^{-1}. Which of the following statements will be true?

(1) The work done in moving a point charge of 1 μC from the given point to a point at a distance of 9 m will be 4×10^{-4} J.

(2) The distance of the given point from the charge is 3 m.

(3) The potential at a distance of 9 m will be 200 V.
(4) The magnitude of charge is 0.2×10^{-3} C.

15. An electric charge 2×10^{-8} C is placed at the point (1, 2, 4). At the point (3, 2, 1), the electric
(1) field will increase by a factor K if the space between the points is filled with a dielectric of dielectric constant K
(2) field will be along the y-axis
(3) potential will be 49.9 V
(4) field will have no y-component

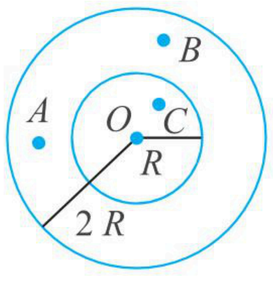
16. The electric potential in a region along the x-axis varies with x according to the relation $V(x)=4 + 5x^2$. Then
(1) potential difference between the points $x = 1$ and $x = -2$ is 15 V
(2) the force experienced by the above charge will be toward the $+ x$-axis
(3) a uniform electric field exists in this region along the $+ x$-axis
(4) force experienced by a 1 C charge at $x = -1$ m will be 10 N

17. A conducting sphere of radius R has a charge. Then,
(1) the charge is uniformly distributed over its surface, if there is no external electric field
(2) the distribution of charge over its surface will be nonuniform if an external electric field exists in the space
(3) the potential at every point of the sphere must be the same
(4) the electric field strength inside the sphere will be equal to zero only when no external electric field exists

18. A small conducting sphere of radius a mounted on an insulated handle and a positive charge q is inserted through a hole in the wall of a hollow conducting sphere of inner radius b and outer radius c. The hollow sphere is supported on an insulating stand and is initially uncharged. The small sphere is placed at the center of the hollow sphere. Neglect any effect of the hole. Which of the following statements will be true for this system?
(1) No work will be done in carrying a small charge from the inner conductor to the outer conductor.
(2) The electric field at a point in the region between the spheres at a distance r from the center is equal to $q / 4\pi\varepsilon_0 r^2$.
(3) The electric field at a point outside the hollow sphere at a distance r from the center is $q / 4\pi\varepsilon_0 r^2$.
(4) The potential of the inner sphere with respect to the outer sphere is given by $V_{ab} = \dfrac{q}{4\pi\varepsilon_0}\left(\dfrac{1}{a} - \dfrac{1}{b}\right)$.

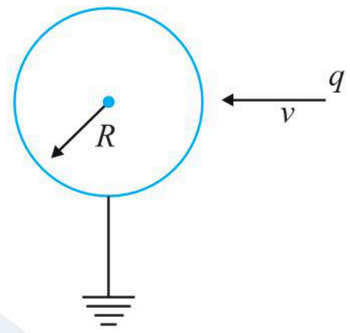
19. Mark the correct statements.
(1) A given conducting sphere cannot be charged to a potential greater than a certain value.
(2) A given conducting sphere can be charged to a potential less than a certain minimum value.
(3) A given conducting sphere can be charged to any extent.
(4) None of the above

20. A hollow conducting sphere of inner radius r and outer radius $2R$ is given charge Q as shown in the figure, then the



(1) potential at A and B is same
(2) potential at O and B is same
(3) potential at O and C is same
(4) potential at A, B, C and O is same

21. A charged particle having a positive charge q approaches a grounded metallic spheres of radius R with a constant small speed v as shown in figure.



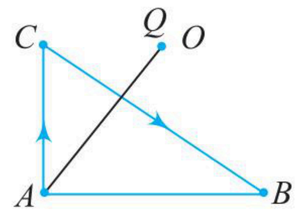
In this situation,
(1) as the charge draws nearer to the surface of the sphere, a current flows into the ground
(2) as the charge draws nearer to the surface of the sphere, a current flows out of the ground into the sphere
(3) as the charged particle draws nearer, the magnitude of the current flowing in the connector joining the shell to the ground increases
(4) as the charged particle draws nearer, the magnitude of the current flowing in the connector joining the sphere to the ground decreases

22. A positive charge having magnitude q is fixed at the origin. An electric dipole having dipole moment $\vec{p}$ is placed along the x-axis at a large distance from the origin with $\vec{p}$ pointing along the positive x-axis and it is released from rest and the dipole starts moving. The kinetic energy of the dipole when it reaches a distance x from the origin is K and the magnitude of the force experienced by charge q at this moment is F. Then
(1) K varies as $1/x$ (2) K varies as $1/x^2$
(3) F varies as $1/x^3$ (4) F varies as $1/x^2$

23. A charged particle X moves directly toward another charged particle Y. For the X and Y system, the total momentum is p and the total energy is E.
(1) If Y is fixed, E is conserved but not p.
(2) If Y is fixed, neither E nor p is conserved.
(3) p and E are conserved if both X and Y are free to move.
(4) (a) is true only if X and Y have similar charges.

24. Consider the following conclusions regarding the components of an electric field at a certain point in space given by $E_x = -Ky$, $E_y = Kx$, $E_z = 0$.
(1) The field is conservative.
(2) The field is nonconservative.
(3) The lines of force are straight lines.
(4) The lines of force are circles.

25. There is a fixed positive charge Q at O, and A and B are points equidistant from O. A positive charge $+ q$ is taken slowly by an external agent from A to B along the line AC and then along the line CB.



(1) The total work done on the charge is zero.
(2) The work done by the electrostatic force from A to C is negative.
(3) The work done by the electrostatic force from C to B is positive.
(4) The work done by electrostatic force in taking the charge from A to B is dependent on the actual path.

26. The electric potential in a region is given by the relation $V(x) = 4 + 5x^2$. If a dipole is placed at position (–1, 0) with

dipole moment $\vec{P}$ pointing along positive y-direction, then

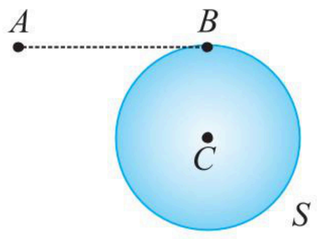
(1) net force on the dipole is zero

(2) net torque on the dipole is zero

(3) net torque on the dipole is not zero and it is in clockwise direction

(4) Net torque on the dipole is not zero and it is in anticlockwise direction

27. S is a solid neutral conducting sphere. A point charge q of 1×10^{-6} C is placed at point, C is the center of sphere, and AB is a tangent. $BC = 3$ m and $AB = 4$ m.



(1) The electric potential of the conductor is 1.8 kV

(2) The electric potential of the conductor is 2.25 kV

(3) The electric potential at B due to induced charges on the sphere is –0.45 kV.

(4) The electric potential at B due to induced charges on the sphere is 0.45 kV.

Linked Comprehension Type

For Problems 1–4

Four charges $+q$, $+q$, $-q$, and $-q$ are placed, respectively, at the corners A, B, C, and D of a square of side a, arranged in the given order. E and F are the midpoints of sides BC and CD, respectively. O is the center of square.

1. The electric field at O is

(1) $\dfrac{q}{\sqrt{2}\pi\varepsilon_0 a^2}$ (2) $\dfrac{q}{\sqrt{3}\pi\varepsilon_0 a^2}$

(3) $\dfrac{\sqrt{3}q}{\pi\varepsilon_0 a^2}$ (4) $\dfrac{\sqrt{2}q}{\pi\varepsilon_0 a^2}$

2. The electric potential at O is

(1) $\dfrac{\sqrt{2}q}{\pi\varepsilon_0 a}$ (2) $\dfrac{\sqrt{3}q}{\pi\,\varepsilon_0 a}$

(3) $\dfrac{q}{\pi\varepsilon_0 a}$ (4) zero

3 The work done in carrying a charge e from O to E is

(1) $\dfrac{\sqrt{2}q\,e}{\pi\,\varepsilon_0 a}$ (2) $\dfrac{q\,e}{\pi\,\varepsilon_0 a}\left[\dfrac{1}{\sqrt{5}}-1\right]$

(3) $\dfrac{q\,e}{\pi\,\varepsilon_0 a}\left[\dfrac{1}{\sqrt{5}}+1\right]$ (4) zero

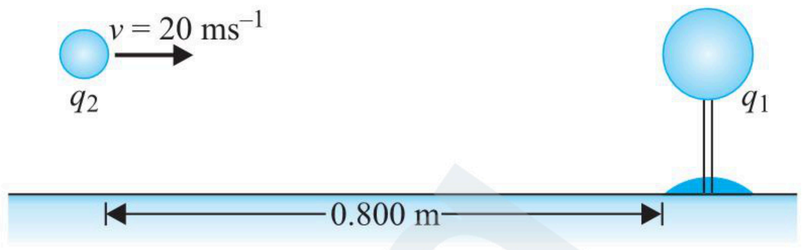
4. The work done in carrying a charge e from O to F is

(1) $\dfrac{\sqrt{2}q\,e}{\pi\,\varepsilon_0 a}$ (2) $\dfrac{q\,e}{\pi\,\varepsilon_0 a}\left[\dfrac{1}{\sqrt{5}}-1\right]$

(3) $\dfrac{q\,e}{\pi\,\varepsilon_0 a}\left[\dfrac{1}{\sqrt{5}}+1\right]$ (4) zero

For Problems 5–6

A small metal sphere, carrying a net charge $q_1 = -2$ μC, is held in a stationary position by insulating supports. A second small metal sphere, with a net charge of $q_2 = -8$ μC and mass 1.50 g, is projected toward q_1. When the two spheres are 0.800 m apart, q_2 is moving toward q_1 with speed 20 ms^{-1} as shown in figure. Assume that the two spheres can be treated as point charges. You can ignore the force of gravity.



5. The speed of q_2 when the spheres are 0.400 m apart is

(1) $2\sqrt{10}$ ms^{-1} (2) $2\sqrt{6}$ ms^{-1}

(3) $4\sqrt{10}$ ms^{-1} (4) $4\sqrt{6}$ ms^{-1}

6. How close does q_2 get to q_1?

(1) 0.20 m (2) 0.30 m

(3) 0.10 m (4) 0.15 m

For Problems 7–8

A charge Q is distributed over two concentric hollow spheres of radii r and R ($R > r$) such that their surface densities are equal.

7. The charges on smaller and bigger shells are

(1) $\dfrac{Qr^2}{r^2+R^2}$ and $\dfrac{QR^2}{r^2+R^2}$, respectively

(2) $Q\left(1+\dfrac{r^2}{R^2}\right)$ and $Q\left(1+\dfrac{R^2}{r^2}\right)$, respectively

(3) $Q\left(1-\dfrac{r^2}{R^2}\right)$ and $Q\left(1-\dfrac{R^2}{r^2}\right)$, respectively

(4) $\dfrac{QR^2}{r^2+R^2}$ and $Q\dfrac{Qr^2}{r^2+R^2}$, respectively

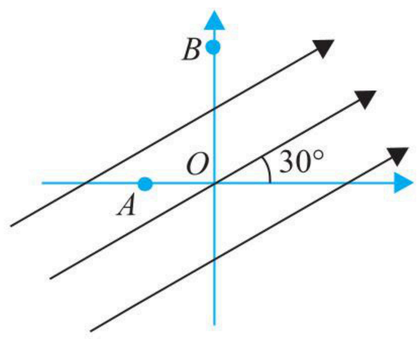
8. The potential at the common centre is

(1) $\dfrac{\sqrt{2}}{\pi\varepsilon_0}\dfrac{Q\,(R+r)}{(R^2+r^2)}$ (2) $\dfrac{1}{2\pi\varepsilon_0}\dfrac{Q\,(R+r)}{(R^2+r^2)}$

(3) $\dfrac{1}{4\pi\varepsilon_0}\dfrac{Q\,(R+r)}{(R^2+r^2)}$ (4) $\dfrac{1}{\pi\varepsilon_0}\dfrac{Q\,(R-r)}{(R^2+r^2)}$

For Problems 9–10

A uniform electric field of 100 Vm^{-1} is directed at 30° with the positive x-axis as shown in figure. $OA = 2$ m and $OB = 4$ m.



9. The potential difference $V_O - V_A$ is

(1) $100\sqrt{3}$ V (2) $200\sqrt{3}$ V

(3) $-100\sqrt{3}$ V (4) $-200\sqrt{3}$ V

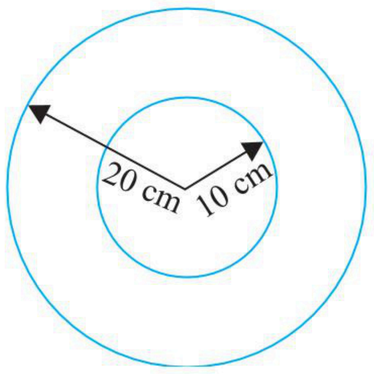
10. The potential difference $V_B - V_A$ is

(1) $-100\left[2+\sqrt{3}\right]$ V (1) $100\left[2+\sqrt{3}\right]$ V

(3) $100\left[2-\sqrt{3}\right]$ V (4) $-100\left[2-\sqrt{3}\right]$ V

For Problems 11–14

We have an isolated conducting spherical shell of radius 10 cm. Some positive charge is given to it so that the resulting electric field has a maximum intensity of 1.8×10^6 NC^{-1}. The same amount of negative charge is given to another isolated conducting spherical shell of radius 20 cm. Now, the first shell is placed inside the second so that both are concentric as shown in figure.



11. The electric potential at any point inside the first shell is

(1) 18×10^4 V (2) 9×10^4 V

(3) 4.5×10^4 V (4) 1.8×10^4 V

12. The electric field intensity just inside the outer sphere is

(1) 4.5×10^5 NC^{-1} (2) 9×10^5 NC^{-1}

(3) 4.5×10^4 NC^{-1} (4) 5×10^4 NC^{-1}

13. The electrostatic energy stored in the system is

(1) 1.0 J (2) 0.045 J

(3) 0.09 J (4) 1.8 J

14. If both the spheres are connected by a conducting wire, then

(1) nothing will happen

(2) some part of the energy stored in the system will convert into heat

(3) charge on both spheres will be positive

(4) entire amount of the energy stored in the system will convert into heat

For Problems 15–18

The electric potential varies in space according to the relation $V = 3x + 4y$. A particle of mass 0.1 kg starts from rest from point (2, 3.2) under the influence of this field. The charge on the particle is +1 μC. Assume V and (x, y) are in SI units.

15. The component of electric field in the x-direction (E_x) is

(1) -3 Vm^{-1} (2) 4 Vm^{-1}

(3) 5 Vm^{-1} (4) 8 Vm^{-1}

16. The component of electric field in the y-direction (E_y) is

(1) 3 Vm^{-1} (2) -4 Vm^{-1}

(3) 5 Vm^{-1} (4) 8 Vm^{-1}

17. The time taken to cross the x-axis is

(1) 20 s (2) 40 s

(3) 200 s (4) 400 s

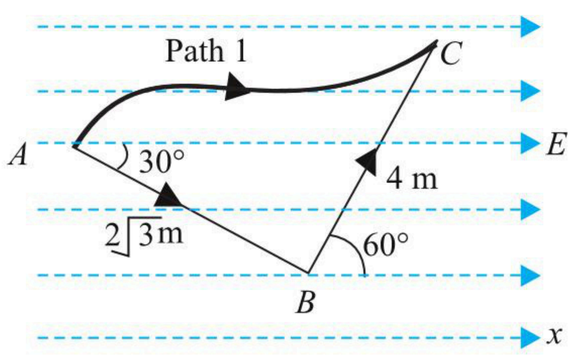
18. The velocity of the particle when it crosses the x-axis is

(1) 20×10^{-3} ms^{-1} (2) 40×10^{-3} ms^{-1}

(3) 30×10^{-3} ms^{-1} (4) 50×10^{-3} ms^{-1}

For Problem 19–21

In a certain region, electric field E exists along the x-axis which is uniform. Given $AB = 2\sqrt{3}$ m and $BC = 4$ m. Points A, B, and C are in xy plane.



19. Find the potential difference $V_A - V_B$ between the points A and B.

(1) $3E$ (2) $4E$

(3) E (4) $2E$

20. Find the potential difference $V_C - V_B$ between the points C and B.

(1) $2E$ (2) $3E$

(3) $-E$ (4) $-2E$

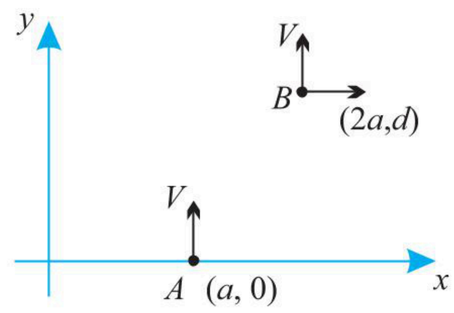
21. A charged particle q is moved from A to C as shown in path 1. What is the work done by electric field in this process?

(1) $2qE$ (2) $5qE$

(3) qE (4) $4qe$

For Problems 22–24

An uniform electric field of strength E exists in a region. An electron of mass m enters a point A perpendicular to x-axis with velocity V. It moves through the electric field and exits at point B. The components of velocity at B are shown in figure. At B the y-component of velocity remain unchanged.



22. Find electric field

(1) $\frac{2maV^2}{ed^2}$ (2) $\frac{maV^2}{ed^2}$

(3) $\frac{maV^2}{2ed^2}$ (4) $\frac{2maV^3}{ed^3}$

23. Find velocity at B

(1) $V\sqrt{1-\left(\frac{2a}{d}\right)^2}$ (2) $V\sqrt{1+\left(\frac{2a}{d}\right)^2}$

(3) $V\sqrt{1+\left(\frac{a}{d}\right)^2}$ (4) $V\sqrt{2+\left(\frac{2a}{d}\right)^2}$

24. Find the rate of work done by the field at B

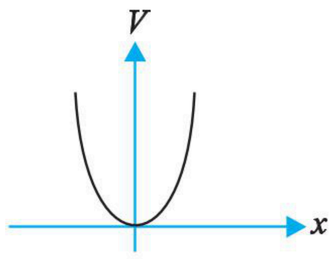
(1) $\frac{2ma^2V^3}{d^3}$ (2) $\frac{ma^2V^3}{2d^3}$

(3) $\frac{4ma^2V^3}{d^3}$ (4) $\frac{ma^2V^3}{d^3}$

For Problems 25–27

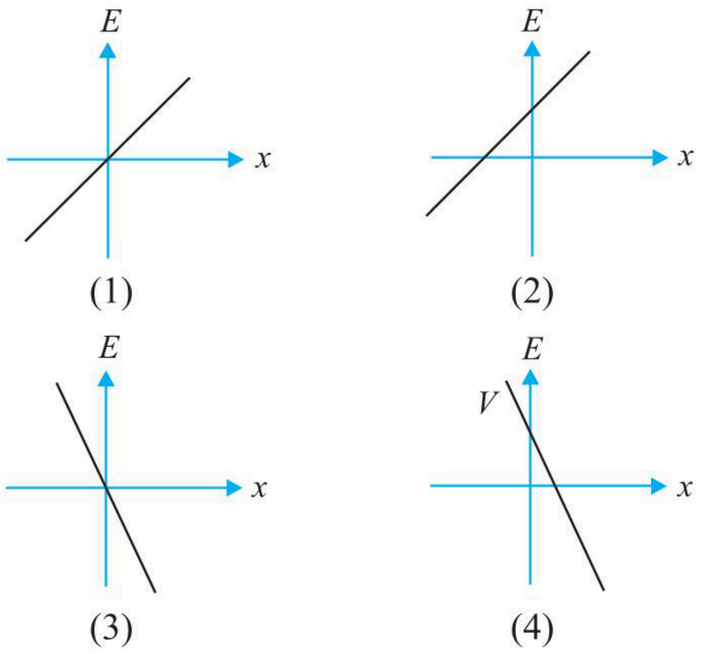
We know that electric field (E) at any point in space can be calculated using the relation

$$\vec{E} = -\frac{\delta V}{\delta x}\hat{i} - \frac{\delta V}{\delta y}\hat{j} - \frac{\delta V}{\delta z}\hat{k}$$



if we know the variation of potential (V) at that point. Now let the electric potential in volt along the x-axis vary as $V = 2x^2$, where x is in meter. Its variation is as shown in figure.

25. A charge particle of mass 10 mg and charge 2.5 μC is released from rest at $x = 2$ m. The variation of electric field (E) along the x-axis is expressed by



26. The velocity of the particle when it crosses origin is

(1) 0.5 ms^{-1} (2) 1 ms^{-1}

(3) 2 ms^{-1} (4) 4 ms^{-1}

27. Will the particle perform a simple harmonic motion? The time period of its oscillations of the particle should be

(1) Yes, time period = 2π s

(2) No, time period = 2π s

(3) Yes, time period = 4π s

(4) The particle will perform SHM, but time period cannot be found from the given data.

For Problems 28–32

A point charge q_1 is placed inside cavity 1 and another point charge q_2 is inside cavity 2. A point charge q is placed outside the conductor. For the situation described above, answer the following questions.

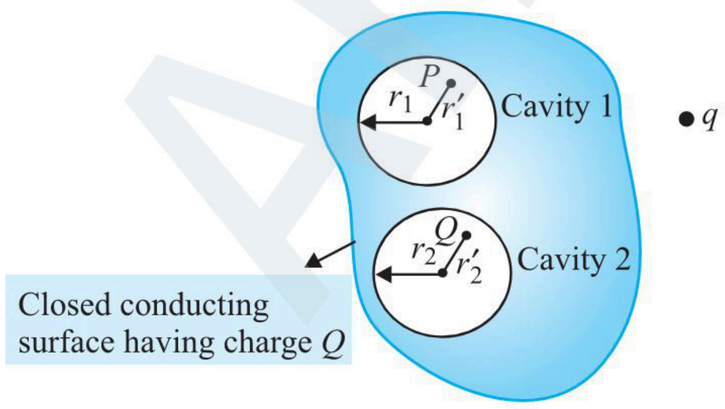
28. The charge on outer surface of the conductor would be

(1) $Q + q_1 + q_2$ and nonuniformly distributed

(2) $Q + q_1 + q_2$ and its uniform or nonuniform distribution depends upon locations of q_1 and q_2

(3) $Q + q_1 + q_2$ and would be distributed uniformly

(4) $Q + q_1 + q_2$ and the distribution depends upon the locations of q_1, q_2, and q



29. If q_1 is at the center of cavity 1, then $\vec{E}$ at point S, at a distant r from the center of cavity 1($r > r_1$), due to the induced charge on the surface of cavity 1 is

(1) $q_1/4\pi\varepsilon_0 r^2$ away from the center of cavity 1

(2) $q_1/4\pi\varepsilon_0 r_1^2$ away from the center of cavity 1

(3) zero

(4) $q_1/4\pi\varepsilon_0 r^2$ toward the center of cavity 1

30. $\vec{E}$ inside the conductor at point S distant r from point charge q, due to charge on outer surface of the conductor, would be

(1) $(Q + q_1 + q_2)/4\pi\varepsilon_0 r^2$ away from charge q

(2) $q/4\pi\varepsilon_0 r^2$ toward charge q

(3) zero

(4) cannot be determined

31. If charge q_2 is at point Q (inside cavity 2), then $\vec{E}$ at the center of cavity 2 due to induced charge on the surface of cavity 2 would be

(1) $q_2/4\pi\varepsilon_0 r_2'^2$ toward q_2 (2) $q_2/4\pi\varepsilon_0 r_2'^2$ away from q_2

(3) zero (4) cannot be determined

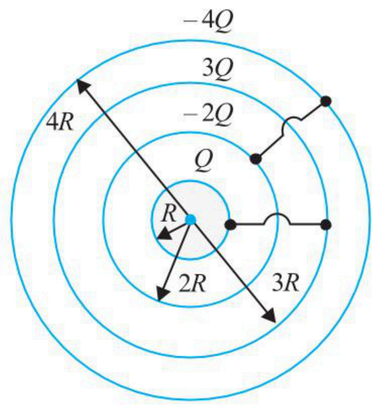
32. If the potential of the conductor is V_0 and charge q_2 is placed at the center of cavity 2, then potential at point Q is

(1) $\dfrac{q_2}{4\pi\varepsilon_0 r_2'} + V_0$ (2) $\dfrac{q_2}{4\pi\varepsilon_0}\left(\dfrac{1}{r_2'} + \dfrac{1}{r_2}\right) + V_0$

(3) $\dfrac{q_2}{4\pi\varepsilon_0}\left(\dfrac{1}{r_2'} - \dfrac{1}{r_2}\right) + V_0$ (4) V_0

For Problems 33–36

Four concentric hollow spheres of radii R, $2R$, $3R$, and $4R$ are given the charges as shown in figure. Then the conductors 1 and 3, 2 and 4 are connected by conducting wires (both the connections are made at the same time).



33. The charge on the inner surface of the third conductor is

(1) $-6Q/5$ (2) $6Q/5$

(3) $-2Q$ (4) $+2Q$

34. The charge on the fourth conductor is

(1) $22Q/5$ (2) $+11Q/3$

(3) $-11Q/3$ (4) $-22Q/5$

35. The potential of conductor 1 is

(1) $3Q/40\pi\varepsilon_0 R$ (2) $-19Q/40\pi\varepsilon_0 R$

(3) $-3Q/40\pi\varepsilon_0 R$ (4) $19Q/40\pi\varepsilon_0 R$

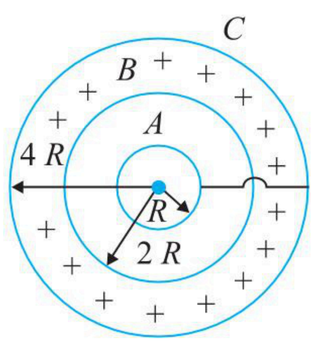
36. The potential of conductor 2 is

(1) $-Q/8\pi\varepsilon_0 R$ (2) $Q/8\pi\varepsilon_0 R$

(3) $Q/32\pi\varepsilon_0 R$ (4) $-Q/32\pi\varepsilon_0 R$

For Problems 37–39

Three concentric spherical conductors A, B, and C of radii R, $2R$, and $4R$, respectively. A and C is shorted and B is uniformly charged (charge+Q).



37. Charge on conductor A is

(1) $Q/3$ (2) $-Q/3$

(3) $2Q/3$ (4) $-2Q/3$

38. Potential at A is

(1) $Q/4\pi\varepsilon_0 R$ (2) $Q/16\pi\varepsilon_0 R$

(3) $Q/20\pi\varepsilon_0 R$ (4) $5Q/48\pi\varepsilon_0 R$

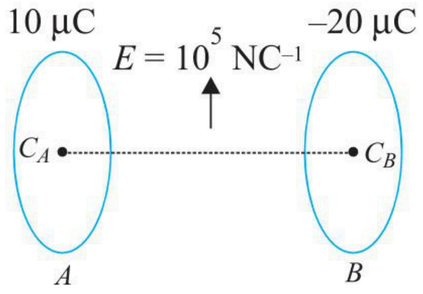
39. Potential at B is

(1) $Q/4\pi\varepsilon_0 R$ (2) $Q/16\pi\varepsilon_0 R$

(3) $Q/48\pi\varepsilon_0 R$ (4) $5Q/48\pi\varepsilon_0 R$

For Problem 40–41

Two circular rings A and B each of radius $a = 30$ cm are placed coaxially with their axis horizontal in a uniform electric field $E = 10^5$ NC^{-1} directed vertically upward as shown in figure. Distance between centers of the rings A and B (C_A and C_B) is 40 cm. Ring A has positive charge $q_A = 10\,\mu$C and B has a negative charge $q_B = -20\,\mu$C. A particle of mass m and charge $q = 10\,\mu$C is released from rest at the center of ring A. If particle moves along C_AC_B, then



40. Work done by electric field, when particle moves from C_A to C_B is

(1) -1.2 J (2) 1.2 J

(3) -3.6 J (4) 3.6 J

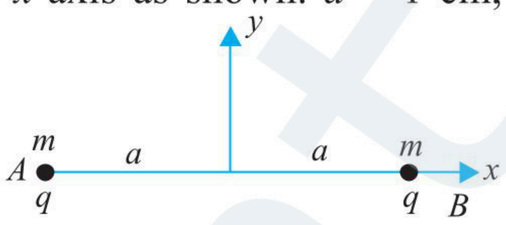
41. Speed of particle when it reaches at center of B is

(1) $6\sqrt{2}$ m/s (2) $12\sqrt{2}$ m/s

(3) $2\sqrt{6}$ m/s (4) $4\sqrt{6}$ m/s

For Problems 42–44

Two point charges are placed on x-axis as shown. $a = 1$ cm, $q = 1\,\mu$C, mass of each particle $m = 6$ g. The charges are tied at the end of an inextensible string of length $2a$. The whole system is free to move on a horizontal frictionless surface.



42. What is the tension in the string?

(1) 11.25 N (2) 22.5 N

(3) 45 N (4) 90 N

43. Now suppose a third charge of equal value $+q$ is fixed at the point $(a/2, 0)$ and the system of charges A and B is released. There is no friction between third charge and string. Find the maximum velocity of the system of charges A and B.

(1) 2 ms^{-1} (2) 10 ms^{-1}

(3) 10 cm s^{-1} (4) 5 ms^{-1}

44. What is the tension in the string when velocity is maximum.

(1) 45 N (2) 67.5 N

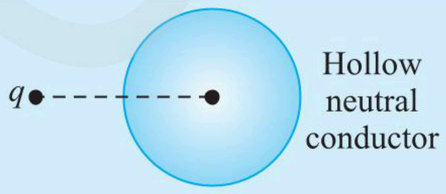
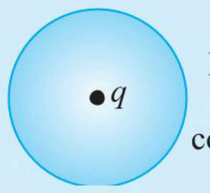
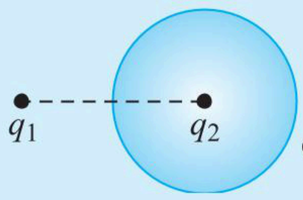
(3) 90 N (4) 112.5 N

Matrix Match Type

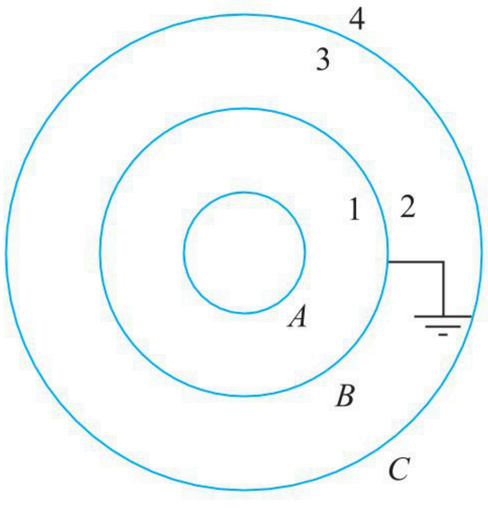
1. Match the columns

Column I	Column II
i. Electrically neutral thick conducting spherical shell, with a point charge at its center.	**a.** Electric potential at the center of the cavity due to charges induced on the inner and outer surfaces of the conductor is zero.
ii. Electrically neutral thick conducting spherical shell, with a point charge inside it and to the right of its center	**b.** Electric field everywhere in the cavity due to charges induced on the inner surface of conductor is zero.
iii. Electrically neutral thick conducting spherical shell, with a point charge at its center. Shell is earthed.	**c.** Electric field everywhere inside the cavity due to charges induced on the outer surface of the conductor is zero or induced charge on the outer surface is zero.
iv. Electrically neutral thick conducting spherical shell, with a point charge inside it and to the right of its center. Shell is earthed.	

2. Match the entries of Column I with the entries of Column II.

Column I	Column II
i. 	**a.** $\vec{E}$ inside the conductor is zero
ii. 	**b.** $\lvert\vec{E}\rvert$ inside the conductor is constant but not zero
iii. 	**c.** $\lvert\vec{E}\rvert$ inside the conductor is varying
	d. Potential inside the conductor is same as that of conductor
	e. Potential inside the conductor varies

3. Figure shows three concentric thin spherical shells A, B, and C of radii R, $2R$, and $3R$, respectively. The shell B is earthed and A and C are given charges q and $2q$, respectively. If the charge on surfaces 1, 2, 3, and 4 are q_1, q_2, q_3, and q_4, respectively, then match the following columns.



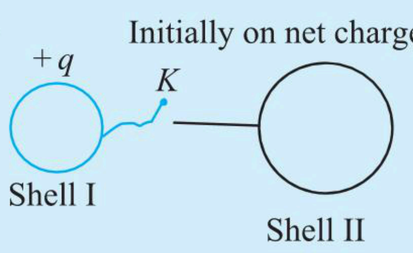
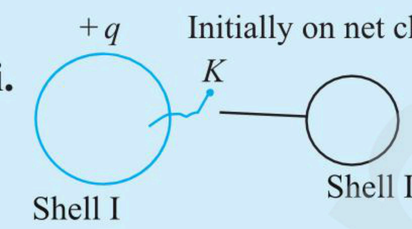
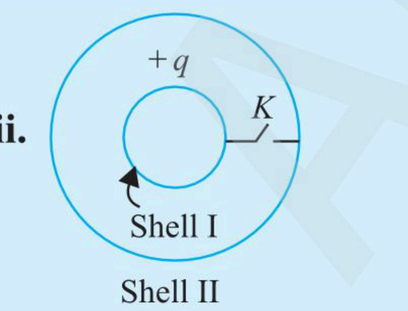
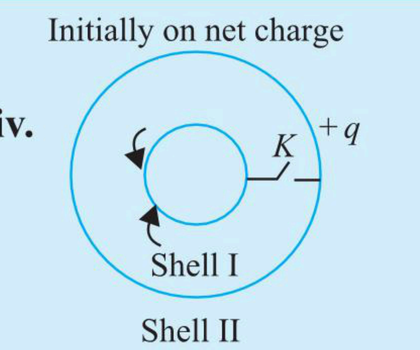
Column I	Column II
i. q_1	**a.** $\frac{2}{3}q$
ii. q_2	**b.** $\frac{4}{3}q$

iii. q_3	**c.** $-\frac{4}{3}q$
iv. q_4	**d.** $-q$

4. A solid conducting sphere of radius a is placed inside a conducting shell of radius b so that both are concentric. Now, the shell is given a charge Q. Match the following.

Column I	Column II
i. Charge appearing on the inner sphere	**a.** zero
ii. Charge appearing on the inner sphere after it is earthed	**b.** Q
iii. Electric field intensity (E) inside the inner sphere (before earthing)	**c.** Qa/b
iv. Electric field intensity (E) inside the inner sphere after it is earthed	**d.** $\frac{1}{4\pi\varepsilon_0}\frac{Q}{ab}$

5. In the following table, Column I gives certain situations involving two thin conducting shells connected by a conducting wire via key K. In all situations, one sphere has net charge $+q$ and other sphere has no net charge. After key K is pressed, Column II gives some resulting effect. Match the figures in Column I with the statements in Column II.

Column I	Column II
i. 	**a.** Charge flows through the connecting wire
ii. 	**b.** Potential energy of system of spheres decreases
iii. 	**c.** No heat is produced
iv. 	**d.** Sphere I has no charge after equilibrium is reached

6. An electric dipole is placed in a uniform external electric field. θ is the angle between the dipole moment and the field direction. In general, the dipole rotates under a torque. With reference to the behavior of the dipole in an electric field, match Column I with Column II.

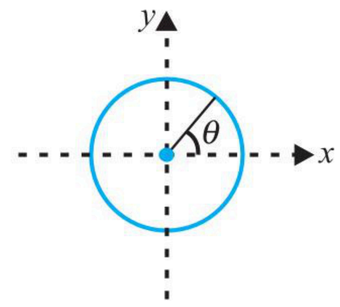
Column I	Column II
i. Potential energy of the dipole is maximum	**a.** conserved
ii. Angular acceleration of the dipole is maximum	**b.** not conserved
iii. Angular momentum of the dipole	**c.** $\theta = 180°$
iv. Kinetic energy of the dipole	**d.** $\theta = 90°$

7. A certain electric field is given as

$$E = -[(2xy + z^2)\hat{i} + (2yz + x^2)\hat{j} + (2zx + y^2)\hat{k}]$$

Column I	Column II
i. Work done by electric field in taking a unit charge from (0, 0, 0) to (3, 4, 0) along a straight line is	**a.** −36 units
ii. Work done by electric field in taking a charge from (3, 4, 0) to (0, 0, 0) along a straight line is	**b.** +36 units
iii. Work done by external agent in taking a charge from (3, 4, 0) to (6, 8, 0) without change in KE is	**c.** zero unit
iv. Net charge enclosed in a sphere of radius 5 units centered at the origin is	**d.** +252 units

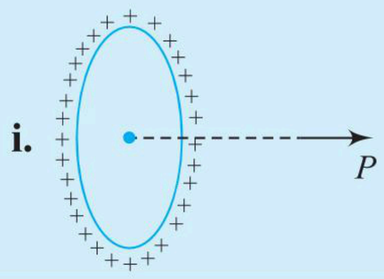
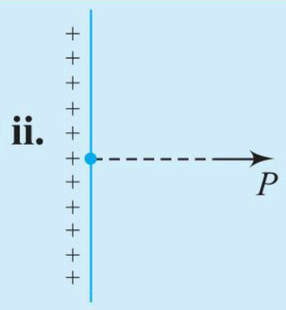
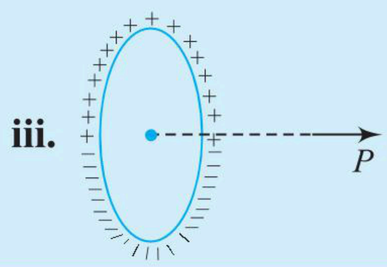
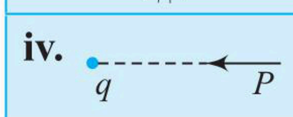
8. A nonconducting ring has linear charge density λ. Match the following column regarding this ring.



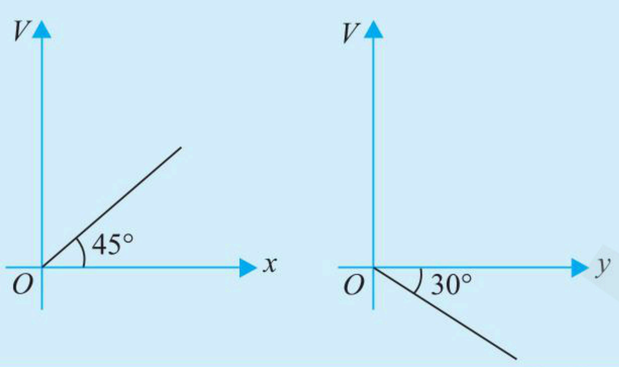
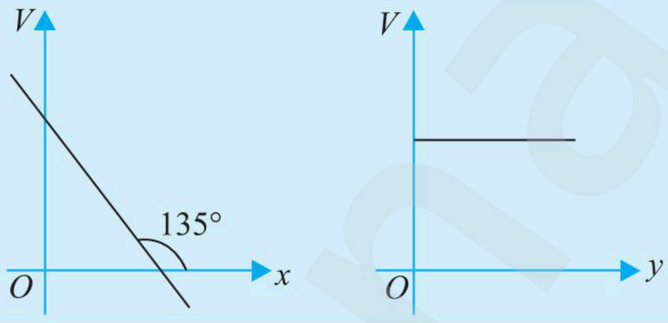
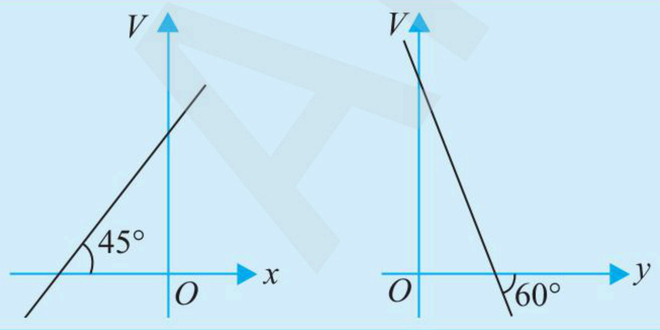
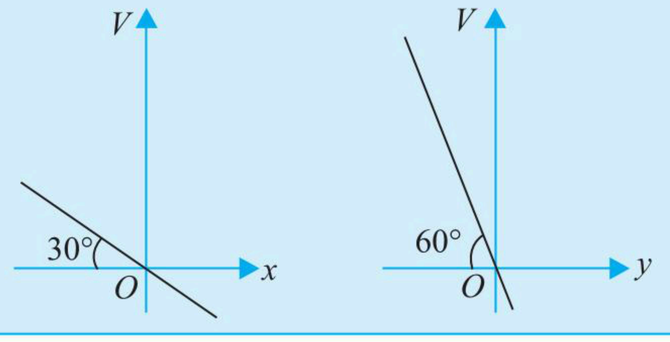
Column I	Column II
i. $\lambda = \lambda_0$ Cm^{-1}	**a.** Electric field at center of the ring is in the direction of $(-\hat{j})$.
ii. $\lambda = \lambda_0 \cos\theta$ Cm^{-1}; $0 \le \theta \le \pi$	**b.** Electric field at center of the ring is zero.
iii. $\lambda = \lambda_0 \sin\theta$ Cm^{-1}; $0 \le \theta \le 2\pi$	**c.** Electrostatic potential at center of the ring is zero.
iv. $\lambda = \lambda_0 (1 - \cos 2\theta)$ Cm^{-1}; $0 \le \theta \le 2\pi$	**d.** Electric field at center of the ring is in the direction $(-\hat{i})$.

9. A short dipole of dipole moment P is kept near different sources of electric field (an infinite line charge, a point charge, a ring charge, etc) in Column I. The force F

experienced by the dipole, the torque t acting on the dipole and the potential energy U of dipole are given in Column II.

Column I	Column II
i. 	a. $F = 0$
ii. 	b. $\tau = 0$
iii. 	c. $F \neq 0$
iv. 	d. $U < 0$

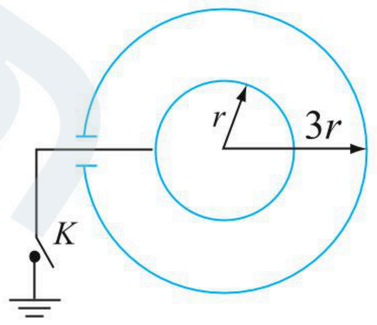
10. Column I shows graphs of electric potential V versus x and y in a certain region for four situations. Column II gives the angle that the electric field vector makes with positive x-direction.

Column I	Column II
i. 	a. 0°
ii. 	b. $\tan^{-1}(3)$
iii. 	c. 120°
iv. 	d. 150°

Numerical Value Type

1. There are 27 drops of a conducting fluid. Each drop has radius r, and each of them is charged to the same potential V_1. They are then combined to form a bigger drop. The potential of the bigger drop is V_2. Find the ratio V_2/V_1. Ignore the change in density of the fluid on combining the drops.

2. Figure shows two conducting thin concentric shells of radii r and $3r$. The outer shell carries charge q and the inner shell is neutral. The amount of charge that flows from the inner shell to the earth after the key K is closed is equal to $(1/n)^{th}$ of the charge on the outer shell. What is the value of n?

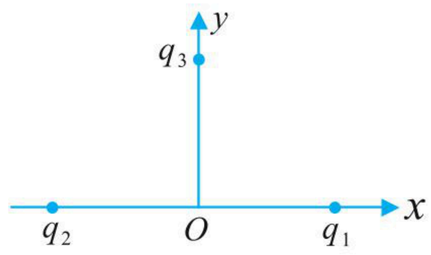


3. A charge of 5 μC is placed at the center of a square $ABCD$ of side 10 cm. Find the work done (in μJ) in moving a charge of 1 μC from A to B.

4. The linear charge density on a dielectric ring of radius R varies with θ as $\lambda = \lambda_0 \cos \theta/2$, where λ_0 is constant. Find the potential at the center O of the ring [in volt].

5. The point charges $-2q$, $-2q$, and $+q$ are put on the vertices of an equilateral triangle of side a. Find the work done by some external force in increasing the separation to $2a$ (in joules).

6. Two points charges $q_1 = q_2 = 2$ μC are fixed at $x_1 = +3$ m and $x_2 = -3$ m as shown. A third particle of mass 1 g and charge $q_3 = -4$ μC is released from rest at $y = 4$ m. Find the speed (in terms of the nearest integer in ms^{-1}) of the particle as it reaches the origin.



7. Two concentric spherical conducting shells of radii R and $2R$ are carrying charges q and $2q$, respectively. Both are now connected by a conducting wire. Find the change in electric potential (in V) on the outer shell.

8. A uniform electric field of 10 NC^{-1} exists in the vertically downward direction. Find the increase in the electric potential (in V) as one goes up through a height of 50 cm.

9. The kinetic energy of a charged particle decreases by 10 J as it moves from a point at potential 100 V to a point at potential 200 V. Find the charge on the particle (in $\times 10^{-1}$ C).

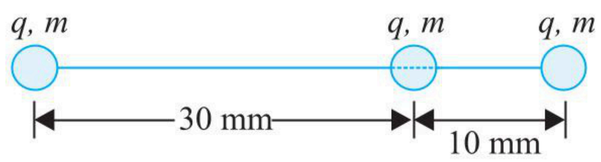
10. The electric field strength depends only on the x, y, and z coordinates according to the law $E = \dfrac{a(x\hat{i} + y\hat{j} + z\hat{k})}{(x^2 + y^2 + z^2)^{3/2}}$, where $a = 280$ Nm^2C^{-1} is a constant. If the potential difference between (3, 2, 6) and (0, 3, 4) is x^2. What is the value of x.

11. A positive charge $+q_1$ is located to the left of a negative charge $-q_2$. On a line passing through the two charges, there are two places where the total potential is zero. The reference is assumed to be at infinity. The first place is between the charges and is 4.00 cm to the left of the negative charge. The

second place is 7.00 cm to the right of the negative charge. If $q_2 = -12\ \mu C$ and $q_1 = 11 \times x\ \mu C$, what is the value of charge x.

12. A positively charged particle starts at rest 25 cm from a second positively charged particle, which is held stationary throughout the experiment. The first particle is released and accelerates directly away from the second particle. When the first particle has moved 25 cm, it has reached a velocity of $10\sqrt{2}\ ms^{-1}$. What is the maximum velocity (in $\times 10\ ms^{-1}$) that the first particle will reach?

13. Two small spheres are attached to the ends of a long light nonconducting rod

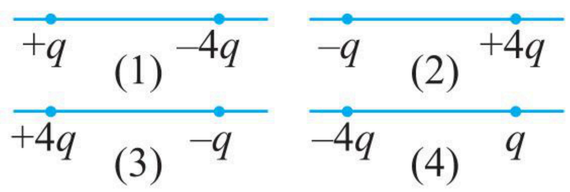


at a distance 40 mm from each other. A third, middle sphere can slide along the rod without friction as shown. All three spheres are nonconducting, have identical masses $m = 1$ g, and a positive charge $q = 1$ C is distributed evenly on the surface of each sphere. The whole system is placed on a horizontal frictionless nonconducting surface. Initially, all three spheres are at rest and the middle sphere is located a distance 30 mm from one of the ends of the rod and a distance10 mm from the other. Find the maximum speed v (in ms^{-1}) of the middle sphere after the system is released.

14. Three point charges of 0.1 C each are placed at the corners of an equilateral triangle with side $L = 1$ m. If this system is supplied energy at the rate of 1 kW, how much time (in hour) will be required to move one of the charges to the midpoint of the line joining the other two?

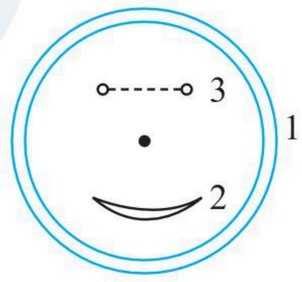
15. A radioactive source in the form of a metal sphere of radius 10^{-2} m emits beta particles (electrons) at the rate of 5×10^{10} particles per second. The source is electrically insulated. How long will it take (in μs) for its potential to be raised by 2 V assuming that 40% of the emitted beta particles escape from the source.

16. The figure shows four situations in which charges as indicated ($q > 0$) are fixed on an axis. How many situations is there a point to the left of the charges where an electron would be in equilibrium?



17. An electric field is given by $\vec{E} = \left(y\hat{i} + x\hat{j}\right)\frac{N}{C}$. Find the work done (in J) in moving a 1C charge from $\vec{r}_A = \left(2\hat{i} + 2\hat{j}\right)$m to $\vec{r}_B = \left(4\hat{i} + \hat{j}\right)$m

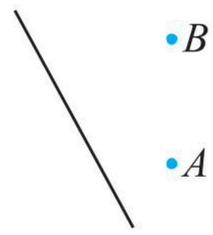
18. The arrangement shown consists of three elements.



- **a.** A thin rod of charge −3.0 μC that forms a full circle of radius 6.0 cm.
- **b.** A second thin rod of charge 2.0 μc that forms a circular arc of radius 4.0 cm and concentric with the full circle, subtending an angle of 90° at the centre of the full circle.
- **c.** An electric dipole with a dipole moment that is perpendicular to a radial line and has magnitude 1.28×10^{-21} C –m

Find the net electric potential in volts at the centre.

19. An infinite plane of charge with $\sigma = 2 \in_0 \frac{C}{m^2}$ is tilted at a 37° angle to the vertical direction as shown below. Find the potential difference, $V_A - V_B$ in volts, between points A and B at 5 m distance apart. (where B is vertically above A).



20. A small sphere of mass m = 0.5 kg carrying a positive charge $q = 110\ \mu C$ is connected with a light, flexible and inextensible string of length $r = 60$ cm and whirled in a vertical circle. If a vertically upward electric field of strength $E = 10^5\ NC^{-1}$ exists in the space, calculate minimum velocity of sphere(in m/s) required at highest point so that it may just complete the circle. ($g = 10\ ms^{-2}$).

Archives

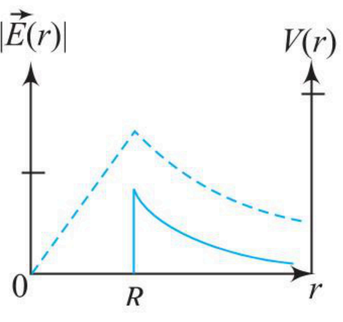
JEE Advanced

Single Correct Answer Type

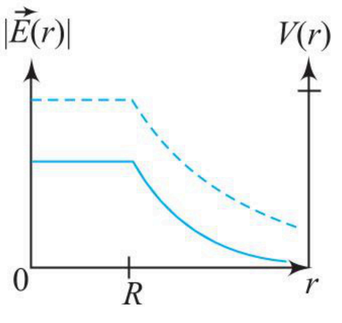
1. Consider a thin spherical shell of radius R with its center at the origin, carrying uniform positive surface charge density. The variation of the magnitude of the electric field $|\vec{E}(r)|$ and the electric potential $V(r)$ with the distance r from the center, is best represented by which graph?

(IIT-JEE 2012)

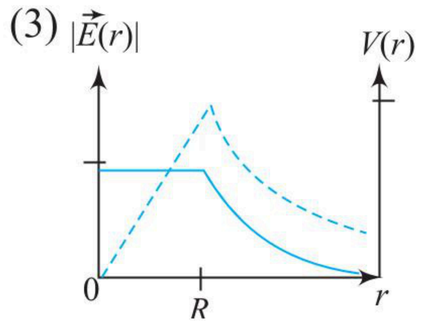
(1)



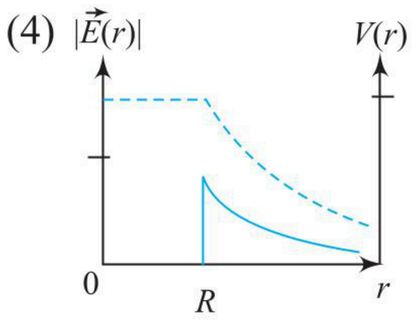
(2)



(3)



(4)



2. Two large vertical and parallel metal plates having a separation of 1 cm are connected to a DC voltage source of potential difference X. A proton is released at rest midway between the two plates. It is found to move at 45° to the vertical just after release. Then X is nearly

(IIT-JEE 2012)

(1) 1×10^{-5} V
(2) 1×10^{-7} V
(3) 1×10^{-9} V
(4) 1×10^{-10} V

3. A thin spherical insulating shell of radius R carries a uniformly distributed charge such that the potential at its surface is V_0. A hole with small area $\alpha 4\pi R^2 (\alpha << 1)$ is made on the shell without affecting the rest of the shell. Which one of the following is correct?

(1) The magnitude of $\vec{E}$ at a point, located on a line passing through the hole and shell's center, on a distance $2R$ from the center of the spherical shell will be reduced by $\frac{\alpha V_0}{2R}$

(2) The potential at the center of shell is reduced by $2\alpha V_0$

(3) The magnitude of electric field at the center of shell is reduced by $\frac{\alpha V_0}{2R}$

(4) The ratio of the potential at the center of the shell to that of the point at $\frac{1}{2}R$ from center towards the hole will be $\frac{1-\alpha}{1-2\alpha}$

(JEE Advanced 2019)

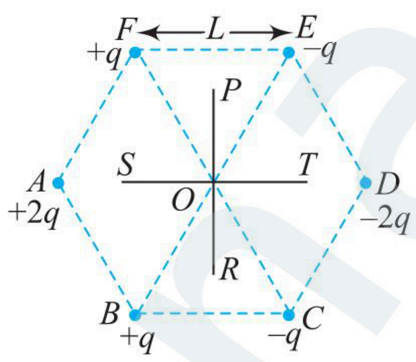
Multiple Correct Answers Type

1. A spherical metal shell A of radius R_A and a solid metal sphere B of radius R_B ($<R_A$) are kept far apart and each is given charge Q. Now they are connected by a thin metal wire. Then **(IIT-JEE 2011)**

(1) $E_A^{\text{inside}} = 0$ (2) $Q_A > Q_B$

(3) $\frac{\sigma_A}{\sigma_B} = \frac{R_B}{R_A}$ (4) $E_A^{\text{on surface}} < E_B^{\text{on surface}}$

2. Six point charges are kept at the vertices of a regular hexagon of side L and center O, as shown in figure. Given that $K = q/4\pi\varepsilon_0 L^2$, which of the following statement(s) is/are correct? **(IIT-JEE 2012)**



(1) The electric field at O is $6K$ along OD.

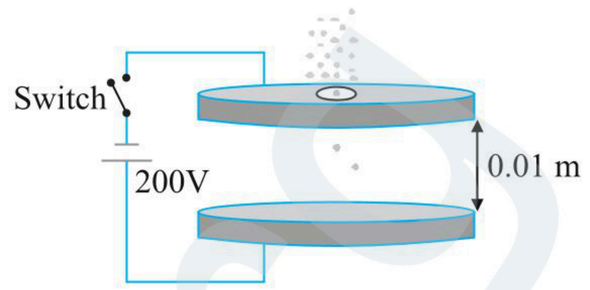
(2) The potential at O is zero.

(3) The potential at all points on the line PR is same.

(4) The potential at all points on the line ST is same.

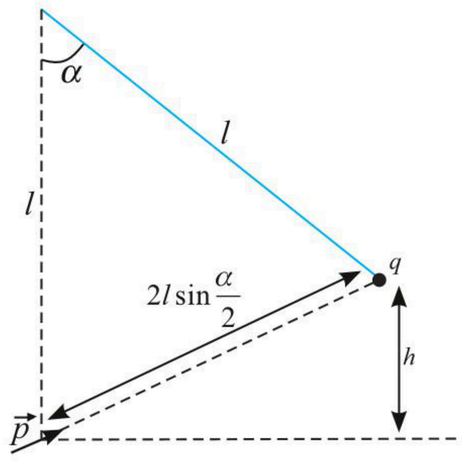
Numerical Value Type

1. Two large circular discs separated by a distance of 0.01 m are connected to a battery via a switch as shown in the figure. Charged oil drops of density 900 kg m^{-3} are released through a tiny hole at the centre of the top disc. Once some oil drops achieve terminal velocity, the switch is closed to apply a voltage of 200 V across the discs. As a result, an oil drop of radius 8×10^{-7} m stops moving vertically and floats between the discs. The number of electrons present in this oil drop is _____ . (neglect the buoyancy force, take acceleration due to gravity = 10 ms^{-2} and charge on an electron (e) = 1.6×10^{-19} C)



(JEE Advanced 2020)

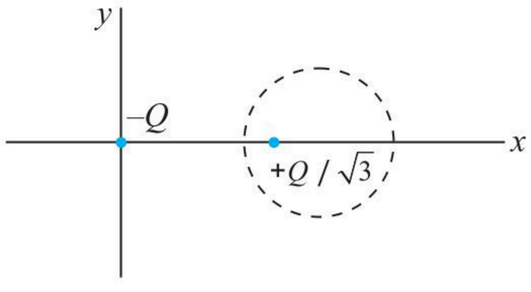
2. A point charge q of mass m is suspended vertically by a string of length l. A point dipole of dipole moment $\vec{p}$ is now brought towards q from infinity so that the charge moves away. The final equilibrium position of the system including the direction of the dipole, the angles and distances is shown in the figure below. If the work done in bringing the dipole to this position is $N \times (mgh)$, where g is the acceleration due to gravity, then the value of N is_____. (Note that for three coplanar forces keeping a point mass in equilibrium, $\frac{F}{\sin\theta}$ is the same for all forces, where F is any one of the forces and θ is the angle between the other two forces)



(JEE Advanced 2020)

Paragraph for Questions 3 and 4:

Two-point charges $-Q$ and $+Q/\sqrt{3}$ are placed in the xy-plane at the origin (0, 0) and a point (2, 0), respectively, as shown in the figure. This results in an equipotential circle of radius R and potential $V = 0$ in the xy-plane with its center at (b, 0). All lengths are measured in meters.



3. The value of R is _____ meter.

4. The value of b is _____ meter. **(JEE Advanced 2021)**

Answers Key

EXERCISES

Single Correct Answer Type

1. (4)	**2.** (2)	**3.** (4)	**4.** (4)	**5.** (4)
6. (2)	**7.** (2)	**8.** (2)	**9.** (1)	**10.** (2)
11. (4)	**12.** (2)	**13.** (1)	**14.** (4)	**15.** (1)
16. (4)	**17.** (3)	**18.** (3)	**19.** (1)	**20.** (3)
21. (4)	**22.** (4)	**23.** (3)	**24.** (3)	**25.** (3)
26. (1)	**27.** (2)	**28.** (2)	**29.** (2)	**30.** (1)
31. (3)	**32.** (1)	**33.** (1)	**34.** (1)	**35.** (1)
36. (1)	**37.** (3)	**38.** (2)	**39.** (4)	**40.** (4)
41. (2)	**42.** (1)	**43.** (1)	**44.** (3)	**45.** (2)
46. (4)	**47.** (3)	**48.** (2)	**49.** (1)	**50.** (1)
51. (3)	**52.** (1)	**53.** (1)	**54.** (1)	**55.** (2)
56. (4)	**57.** (1)	**58.** (1)	**59.** (3)	**60.** (2)
61. (3)	**62.** (3)	**63.** (2)		

Multiple Correct Answers Type

1. (1),(4)	**2.** (1),(4)
3. (1),(2),(3)	**4.** (1),(4)
5. (2),(4)	**6.** (1),(4)
7. (1),(2),(3),(4)	**8.** (1),(3)
9. (1),(3),(4)	**10.** (1),(2),(3)
11. (1),(3)	**12.** (1),(2)
13. (2),(3),(4)	**14.** (1),(2),(3)
15. (3),(4)	**16.** (1),(3),(4)
17. (1),(2),(3)	**18.** (2),(3),(4)
19. (1),(2)	**20.** (1),(2),(3),(4)
21. (1),(3)	**22.** (2),(3)
23. (1),(3)	**24.** (2),(4)
25. (1),(2),(3)	**26.** (1),(3)
27. (1),(3)	

Linked Comprehension Type

1. (4)	**2.** (4)	**3.** (4)	**4.** (2)	**5.** (3)
6. (2)	**7.** (1)	**8.** (3)	**9.** (3)	**10.** (1)
11. (2)	**12.** (1)	**13.** (3)	**14.** (4)	**15.** (1)
16. (2)	**17.** (4)	**18.** (1)	**19.** (1)	**20.** (4)
21. (2)	**22.** (1)	**23.** (2)	**24.** (3)	**25.** (3)
26. (3)	**27.** (1)	**28.** (1)	**29.** (4)	**30.** (2)
31. (4)	**32.** (3)	**33.** (2)	**34.** (4)	**35.** (3)
36. (2)	**37.** (2)	**38.** (2)	**39.** (4)	**40.** (4)
41. (1)	**42.** (2)	**43.** (2)	**44.** (4)	

Matrix Match Type

1. i. → b., c.; ii. → c.; iii. → b., c.; iv. → c.
2. i. → a., d.; ii. → c., e.; iii. → c., e.
3. i. → d.; ii. → c.; iii. → b.; iv. → a.
4. i. → a.; ii. → c.; iii. → a.; iv. → a.
5. i. → a., b.; ii. → a., b.; iii. → a., b., d.; iv. → c., d.
6. i. → c.; ii. → d.; iii. → b.; iv. → b.
7. i. → a.; ii. → b.; iii. → d.; iv. → c.
8. i. → b.; ii. → c, d.; iii. – a, c.; iv. → b.
9. i. → b, c, d.; ii. → b, c, d.; iii. → c.; iv. → c.
10. i. → d.; ii. → a.; iii. → c.; iv. → b.

Numerical Value Type

1. (9)	**2.** (3)	**3.** (0)	**4.** (0)	**5.** (0)
6. (6)	**7.** (0)	**8.** (5)	**9.** (1)	**10.** (4)
11. (4)	**12.** (2)	**13.** (20)	**14.** (50)	**15.** (700)
16. (2)	**17.** (0)	**18.** (0)	**19.** (3)	**20.** (6)

ARCHIVES

JEE Advanced

Single Correct Answer Type

1. (4) **2.** (3) **3.** (4)

Multiple Correct Answers Type

1. (1),(2),(3),(4) **2.** (1),(2),(3)

Numerical Value Type

1. (6) **2.** (2) **3.** (1.73) **4.** (3.00)

Ananth